Erdős Problem #1088 (JSP-000905)

Candidate proof that $f_d(n) = 2^{O_n(\sqrt{d \log d})} = 2^{o(d)}$ for every fixed n

Compiled 2026-11-10 from the AI research notes for this project. Research record, not the manuscript.

Verification status. This is a CANDIDATE proof developed by AI agents (Claude, Anthropic) in sessions directed by Will Howe. It has passed three independent AI referee passes (Appendices K–M), all reporting "correct" or "correct with fixable gaps", the gaps being expository. **No human expert has reviewed it.** The weakest link is the new Theorem LR (Appendix A), which appears to improve a bound that Singer–Sudan (arXiv:2101.09592) describe as best known; it needs an expert check. The expository referee fixes have not yet been folded into the text. This document is a compilation of working notes, not a polished manuscript, and the notation is plain-text.

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Main proof: f_d(n) = 2^{O_n(sqrt d log d)} for every fixed n (round 14, assembled)

Source file: e1088/round14/general_n_full_proof.md

f_d(n) = 2^{O_n(sqrt d log d)} for every fixed n: assembled proof (round 14)

STATUS: CANDIDATE PROOF, complete in the sense that every step is written out below and labelled; adversarially SELF-checked (05_selfcheck.md) but NOT independently refereed. It is CONDITIONAL on in-project results that have had AI referee passes only (Theorem LR, G1, Theorem R, Theorem R+); see the dependency table. Do not treat as established.

Main Theorem

For every fixed n >= 3 there is C_n such that every set of more than 2^{C_n sqrt(d) log2(d+2)} points of R^d contains n points with all C(n,2) distances distinct. Hence f_d(n) = 2^{O_n(sqrt d log d)} = 2^{o(d)}, answering Erdos #1088 (JSP-000905) affirmatively, conditional as stated above. n = 3: Blokhuis (CITED). n = 4: round11/n4_full_proof or Theorem R+ (round12/03). n >= 5: this file (n = 5 also round13/01; the Core Lemma below with r = 0, p = 1 is that argument).

Dependencies

label	statement	source	status
LR	rank-r matrix covered by k row/column/constant types => single-type rectangle, sides >= 2^{-L_k(r)}, L_k(r) = O(k^3 sqrt(r+2) log(r+2))	round11/01_label_rounding.md	PROVED in-project (AI referee)
G1	rainbow (n-2)-set T, Y in T-generic points: ordered label-disjoint pairs in Y <= 4^{n-2} mu_n	Y	; degree form
GL	rank <= R', <=	A	
HP	rank(A o B) <= rank A rank B	elementary	PROVED
OR	orthogonal split => min(	R	,
H, CW, DIM	Lemma H (symmetric), Caro-Wei, dimension of lifted polynomials	round8; classical; elementary	PROVED / CITED
Konig	max matching = min vertex cover in finite bipartite graphs	classical	CITED
R+	Claim(j) for j >= n-4 (levels needing no E1)	round12/03 + REFEREE_R.md fixes F1-F3	PROVED in-project (AI referee); single-cap restatement: Thm 1.3 below
R-step	kill polynomial + Caro-Wei + Lemma H at one level	round12/01 Theorem R	PROVED in-project (referee12 "correct")
halving	s'(D) <= A(D)(s'(floor(D/2))+1), A = 2^{C sqrt D log D} => s'(d) = 2^{O(sqrt d log d)}	round11/n4_full_proof Sec. 9	PROVED

Structure

Downward induction over levels r = n-5, ..., 0: Claim(r+1) => HR_r (Lemma 1.1) => E1_r for EVERY column (Theorem 1.4: Core Lemma (Part 2) after n-5-r slicing rounds (Part 3)) => Claim(r) (Lemma 1.2). Levels >= n-4: Theorem R+. Part 4 (= 04_n6.md) is the n = 6 instance; Part 5 summarises the adversarial check (05_selfcheck.md).

Part 1

01. Framework: configurations, Claim(j), HR_r, the level induction, the threshold tower

STATUS: Lemmas 1.1, 1.2, Cor 1.5 PROVED (new write-up of round12/round13 arguments in a "configuration" language). Theorem 1.3 CITED (Theorem R+ of round12/03 with referee fixes F1-F3, restated with a single cap; see the translation remark). Theorem 1.4 (E1_r) is proved in 02_core_lemma.md + 03_slicing.md.

Setting

n >= 5 fixed. X rainbow-n-free, affine dimension <= D. s' := s'_n(floor(D/2)) (largest rainbow-n-free set of affine dimension <= floor(D/2)). mu := mu_n (G1 of round12/01). L_k := L_k(D+2) from Theorem LR (round11/01); L_k may be taken nondecreasing in k (a cover by k types is a cover by k+1 types). For a finite rainbow set A: D_A := its C(|A|,2) distances. S(c,t) sphere, H(p,q) perpendicular bisector. Tools GL, LR, OR, HP, H, CW, DIM, G1 as in round11/n4_full_proof Section 1 and round12/01 "Tools".

Definition (level-j configuration)

A triple (S, W, sigma): S in X rainbow, |S| = j; W in X \ S; S u {x} rainbow for every x in W; and the CAP condition |W n S(s,t)| < sigma for all s in S, t > 0. (For j = 0 the cap condition is void; sigma is a scale.) For b in W: R_b := D_S u {|bs| : s in S} = D_{S u {b}}. For tau > 0: Big_b(tau) := {lam notin R_b : |W n S(b,lam)| >= tau}.

Claim(j) [constant Gamma_j]: every level-j configuration with sigma >= sigma_min has |W| <= Gamma_j sigma.

Lemma 1.1 (HR_r from Claim(r+1); bisector exception) [PROVED]

Assume Claim(r+1). Let (S,W,sigma) be a level-r configuration, sigma >= sigma_min, c in W, V in W \ {c}, theta >= sigma. Then, with Gamma'*r := Gamma*{r+1} + r^2, #{ x in V : |xc| notin R_c, x notin U_{s in S} H(c,s), |V n S(c,|xc|)| < theta } <= Gamma'*r theta. Proof. Let V_0 be the set counted. V_2 := V_0 n U*{s,s' in S} S(s,|cs'|): < r^2 sigma <= r^2 theta points (cap). V_1 := V_0 \ V_2. For x in V_1, S u {c,x} is

rainbow: its distances are D_S, |cs|, |xs|, |xc|; D_S u {|cs|} and D_S u {|xs|} are rainbow (c, x in W); |xs| != |cs'| (x notin V_2); |xc| notin D_S u {|cs|} = R_c; |xc| != |xs| (x notin H(c,s)); x != c. [COMPUTER check "HR" in code/check_core.py: the equivalence holds in all samples.] So (S u {c}, V_1, theta) is a level-(r+1) configuration: S u {c} rainbow; V_1 in X \ (S u {c}); classes of s meet V_1 (in W) in < sigma <= theta points; a class S(c,t) meeting V_1 at x has |V_1 n S(c,t)| <= |V n S(c,|xc|)| < theta. theta >= sigma >= sigma_min, so Claim(r+1) gives |V_1| <= Gamma_{r+1} theta. [] Remark (bisector exception). Points of V on H(c,s) are NOT controlled. Every use below treats x in H(c,s), i.e. |xc|^2 = |xs|^2, as an LR ROW type of the matrix [|xc|^2] (rows x, columns c); a single-type rectangle of this type has R in H(s,c) for all columns c, which gives (OR). Spheres S(c, R_c) are spheres about c with radii |cs| (column functions) or d in D_S (constants).

Lemma 1.2 (kill step at level r, case B_K) [PROVED; = step of Theorem R / case B of R+]

Assume Claim(r+1). Let (S,W,sigma) be a level-r configuration, sigma >= sigma_min, tau >= sigma, and suppose |Big_b(tau)| <= K for EVERY b in W. Then |W| <= P(K,r) (2 Gamma_{r+1} tau + 2 r^2 sigma + 1), P(K,r) := C(D+1+e, e)^2, e := K + 2r + C(r,2). Proof. F_b(y) := prod_{lam in Big_b}(|y-b|^2 - lam^2) prod_s (|y-b|^2 - |y-s|^2) prod_s (|y-b|^2 - |bs|^2) prod_{d in D_S} (|y-b|^2 - d^2): degree <= e in the lifted coordinates (DIM), F_b(b) != 0. If x in W \ {b} and F_b(x) != 0 then |xb| notin Big_b u R_b, |xb| != |xs|. Either |xs| = |bs'| for some s, s' (x in a class of s: < r^2 sigma such x), or S u {b,x} is rainbow (as in 1.1) and x in W' := {x in W \ {b} : S u {b,x} rainbow, |xb| notin Big_b}. (S u {b}, W', tau) is a level-(r+1) configuration (a class of b meeting W' has radius notin R_b u Big_b, so it meets W in < tau points), so |W'| <= Gamma_{r+1} tau. Unkilled graph, Caro-Wei, Lemma H exactly as in round12/01 Theorem R. []

Theorem 1.3 (levels j >= n-4: Theorem R+) [CITED: round12/03 + referee12 fixes F1-F3]

Let Q := (Co n (D+2)^{k'})^{12}, k' := C(n-2,2) + 2n, as in R+. Define Gamma_{n-2} := 4(n-2)^2 and for j = n-3, n-4: Gamma_j := P_j^+ (2 Q Gamma_{j+1} + 2(j^2 + 2n) + 1), P_j^+ := C(D+3+2j+C(j,2), 2+2j+C(j,2))^2. If sigma_min >= sigma_+ := max(tau_min^{R+}, e_A, 24^{n-2} mu + 4) then Claim(j) holds for j = n-2, n-3, n-4. Translation remark (why the citation applies). R+ is proved for W_S built recursively with caps tau_{k-1} for s_k and thresholds tau_r = Q tau_{r-1}. Its proof uses about W_S only: (a) S u {x} rainbow for x in W_S; (b) every class of every s in S meets W_S in < tau_{r-1}; (c) C_i, C_j, A_0 in W_S \ {b} (referee12 item 4: "levels never use points outside the ground set"; item 2 "general-n HH: at levels >= n-4 only the tau_{r-1} bound on the classes of S is used"). With a single cap sigma for all of S and tau_r := Q sigma these hold verbatim: base (degrees < (n-2) sigma), case B (r^2 sigma), Step A (spread: sigma <= tau_r/Q <= |C_i|/Q; S'-coincidences n(n-4) sigma (1/|P|+1/|Q|) <= 2n^2 Q^{-1/2} for |P|,|Q| >= Q^{-1/2} tau_r (F1); G1 term 24^{n-2} mu Q^{1/2}/tau_r with tau_r >= sigma_min

= tau_min^{R+}). The new point b gets cap tau_r = Q sigma, whence the recursion for Gamma_j (e_A <= sigma).

Theorem 1.4 (E1_r, relative, EVERY column) [PROVED in 02 + 03, given HR_r]

For 0 <= r <= n-5 put K_r := 12 * 6^{n-5-r}. Assume HR_r (Lemma 1.1) with constant Gamma'_r. There is ratio_r = (D+2)^{O_n(1)} Gamma'_r^{O(n)} (explicit in 03, eq. (ratio)) such that, if sigma_min >=

sigma_abs (03), every level-r configuration with sigma >= sigma_min satisfies |Big_b(ratio_r sigma)| < K_r for every b in W.

Corollary 1.5 (the induction closes) [PROVED given 1.3, 1.4]

Downward over r = n-5, ..., 0: Claim(r+1) => HR_r (1.1) => E1_r (1.4) => Claim(r) (1.2 with K = K_r - 1, tau = ratio_r sigma), with Gamma_r := P(K_r - 1, r) (2 ratio_r Gamma_{r+1} + 2r^2 + 1). Threshold tower (no circularity): the constants are defined in the order Gamma_{n-2}, Gamma_{n-3}, Gamma_{n-4} (R+); for r = n-5..0: Gamma'*r* = *Gamma*{r+1} + r^2 -> ratio_r -> Gamma_r; finally sigma_min := max(sigma_+, sigma_abs). ratio_r depends only on n, D, Gamma'_r; Gamma_r only on n, D and higher levels; sigma_min only on n, D, s', mu, L *(absolute requirements: mu, e := 2^{L+1}(s'+1), e_A, tau_min^{R+}, 2*4^{n-2}mu+4, all <= sigma_min). Hence all Gamma_r, ratio_r are (D+2)^{c_n} with c_n depending only on n, and sigma_min = 2^{O_n(sqrt D log D)} (s'+1).

Theorem 1.6 (main) [PROVED, conditional on the cited in-project results]

Claim(0) with W = X, sigma = sigma_min gives |X| <= Gamma_0 sigma_min. Hence s'*n(D) <= 2^{C_n sqrt(D) log(D+2)} (s'_n(floor(D/2)) + 1), and the halving recursion (round11/n4_full_proof Section 9) gives f_d(n) = s'_n(d) + 1 = 2^{O_n(sqrt d log d)}. Loss per halving: Gamma_0 = (D+2)^{c_n} (polynomial) times sigma_min / (s'+1) = 2^{O(L) + O_n(log D)}, L = L*{k}(D+2) = O(k^3 sqrt(D+2) log(D+2)) with k* = O(6^n) types (03): 2^{O_n(sqrt D log D)} per halving.

Part 2

02. Core lemma: a rainbow family with n-4-r centres has at most 11 big classes

STATUS: PROVED (new write-up; generalises round13/01 = the case r = 0, p = 1, n = 5). Uses HR_r (01, Lemma 1.1), G1, GL, LR, OR, HP. Elementary inclusions COMPUTER-checked (code/check_core.py: checks G, S1, S1b, S5). This file is "level n-5 in full" (p = 1: B = {b}, S = r spread points) and also the last step at every lower level.

Setting

Level-r configuration (S, W, sigma) (01), 0 <= r <= n-5, sigma >= sigma_min; HR_r holds with constant Gamma'. For c in W: R_c = D_S u {|cs| : s in S}.

Definition (rainbow family at level r)

B in X with p := |B| >= 1; finite index set I; for k in I a nonempty C_k in W and beta(k) = (beta_1(k),...,beta_p(k)) such that (F1) S u B is rainbow; (F2) |x b_i| = beta_i(k) for all x in C_k (b_1..b_p the points of B); (F3) the p|I| numbers beta_i(k) are pairwise distinct and none lies in D_{S u B}; (F4) S u B u {x} is rainbow for every x in C_k; (F5) |xs| notin beta(I) := {beta_i(k') : i <= p, k' in I} for

every x in U_k C_k and s in S. M(F) := min_k |C_k|. The C_k are pairwise disjoint (beta_1) and disjoint from S u B. Passing to I' in I and to nonempty C'_k in C_k preserves (F1)-(F5). beta(k) also denotes the SET of its p values.

Constants (r, n fixed; p := n-4-r)

t := C(n-2,2); w1 := (n-2) + t + p; w2 := t + p; k3(w) := w + 1 + r; k4(w) := w + 2r + C(r,2); k5 := C(r+p,2) + 2p + 2r; k_c := k3(w1) + k4(w1) (k3(w2) <= k3(w1), k4(w2) <= k4(w1), k5 <= k3(w1)). L *:= L_{k}* (D+2) with k >= *k3(w1), k4(w1), k5 (fixed in 03); e := 2^{L+1}*(s'+1). theta_rel(w) := 1/(32 Gamma' (D+2)^{k4(w)}); eps := theta_rel(w2) / (96 (D+2)^{k3(w2)}); M_core := 2^{20} 4^n n^2 Gamma'^2 (D+2)^{2 k_c} sigma. Absolute requirement (part of sigma_abs in 03): sigma_min >= max(mu, e, 2^{L*+1}(s'+1) + 2).

Core Lemma

If p = n-4-r, there is no rainbow family at level r with |I| >= 12 and M(F) >= M_core.

Proof

Suppose F is one; restrict to |I| = 12; M := M(F). For distinct l, m in I, a1 in C_l, a2 in C_m put T := S u B u {a1, a2} (n-2 points, since r + p = n-4). For x notin T: L_x := {|xt| : t in T}; for x in C_k: L_x = {|xs| : s in S} u beta(k) u {|x a1|, |x a2|}. Y_T := T-generic points (x notin T, T u {x} rainbow). Hyp(a1,a2) := U_{s in S, i = 1,2} H(s, a_i) (empty if r = 0). By (F2), when T is rainbow: D_T = D_{S u B} u {|a_i s|} u beta(l) u beta(m) u {|a1 a2|}. (k,k') is VALID for (a1,a2): k != k' in I \ {l,m}, T rainbow and (beta(k) u beta(k')) n D_T = {}.

Step 0 (non-generic loci) [PROVED; COMPUTER check G]

T rainbow, k in I \ {l,m}, beta(k) n D_T = {}, x in C_k \ Y_T => x in S(a1, D_T u beta(k)) u S(a2, D_T u beta(k)) u H(a1,a2) u Hyp(a1,a2) u Sm(a1,a2), Sm(a1,a2) := U_{s in S} S(s, {|a_i s'| : i, s'} u {|a1 a2|}), |C_k n Sm| < r(2r+1) sigma. Proof. x notin T (disjointness). A coincidence among D_T u L_x: |xs| in D_{S u B} u beta(k) or |xs| = |xs'| is impossible (F4); |xs| in beta(l) u beta(m) impossible (F5); |xs| in {|a_i s'|, |a1a2|}: Sm; |xs| = |x a_i|: Hyp; beta_i(k) in D_T excluded; beta_i(k) = beta_j(k) (F3) or = |xs| (F4) impossible; |x a_i| in D_T u beta(k): spheres; |x a1| = |x a2|: H(a1,a2). Each sphere of s meets W in < sigma points (cap); r(2r+1) spheres. []

Step 1 (trichotomy) [PROVED; COMPUTER checks S1, S1b]

Let (k,k') be valid. P := C_k n Y_T, Q := C_k' n Y_T, Q' := {w in Q : L_w n beta(k) = {}}. (i) |Q'| >= eps |C_k'|. By G1 (P u Q in Y_T; P, Q disjoint) the ordered label-disjoint pairs in Q' x P number <= 4^{n-2} mu (|P| + |Q|), so some w0 in Q' has <= ex := 4^{n-2} mu (|C_k| + |C_k'|)/(eps |C_k'|) label-disjoint partners in P. Any other x in P has L_x n L_w0 != {}; as beta(k) n L_w0 = {}, either |x a_i| in L_w0, or |xs| in L_w0 \ beta(k') (F5), i.e. x in S(s, {|w0 s'|} u {|w0 a1|, |w0 a2|}) (r(r+2) spheres of s). With Step 0: C_k in S(a1,W) u S(a2,W) u H(a1,a2) u Hyp(a1,a2) u Ex, W := L_w0 u D_T u beta(k), |W| <= w1, |Ex| <= ex + r(r+2) sigma + r(2r+1) sigma <= delta1 |C_k|, delta1 := 2*4^{n-2} mu/(eps M) + 3 n^2 sigma / M. DEFINE Good_k(a1,a2): C_k is covered as displayed for SOME set W of <= w1 reals and some Ex, |Ex| <= delta1 |C_k|. (ii) |Q'| < eps |C_k'|. If w in Q \ Q' then |w a_i| in beta(k) for some i (|ws| notin beta(k) by F5, beta(k') n beta(k) = {}). With Step 0 for k' and Lam_k'(a1,a2) := S(a1, D_T u beta(k')) u S(a2, D_T u beta(k')) u H(a1,a2) u Hyp(a1,a2): | C_k' \ (Lam_k' u S(a1, beta(k)) u S(a2, beta(k))) | < eps |C_k'| +

r(2r+1) sigma <= 2 eps |C_k'| (ii)*{k,k'} [r(2r+1) sigma <= 2n^2 sigma <= eps M by (P2)]. So for every valid (k,k'): Good_k(a1,a2) or (ii)*{k,k'}.

Step 2 (summing over k) [PROVED]

Fix a1, a2, k'. If J in I \ {l,m,k'}, |J| >= 3, every (k,k') with k in J valid and (ii)_{k,k'}, then |C_k' \ Lam_k'(a1,a2)| < 6 eps |C_k'|. Proof. c := |C_k' \ Lam_k'|; assume c >= 2 eps |C_k'|. For k in J more than c - 2 eps |C_k'| points of C_k' \ Lam_k' lie in S(a1, beta(k)) u S(a2, beta(k)). By (F3) the radius sets beta(k), k in J, are pairwise disjoint, so the sets S(a1, beta(k)) (k in J) are pairwise disjoint, and likewise about a2: the sum over J is <= 2c. So |J| (c - 2 eps |C_k'|) < 2c, c (|J| - 2) < 2 |J| eps |C_k'|, c < 6 eps |C_k'|. [] DEFINE Cov_k'(a1,a2): |C_k' \ Lam_k'(a1,a2)| < 6 eps |C_k'|. (Shape of Good with W = D_T u beta(k'), |W| <= w2.)

Step 3 (two-centre lemma, with spread points) [PROVED]

Let j in I \ {l,m}, a2 in C_m FIXED, w in {w1, w2}, Z in C_l, and for each a1 in Z a set W(a1) of <= w reals and Ex(a1) in C_j, |Ex(a1)| <= delta |C_j|, delta <= theta_rel(w)/(16 (D+2)^{k3(w)}), with C_j in S(a1, W(a1)) u S(a2, W(a1)) u H(a1,a2) u Hyp(a1,a2) u Ex(a1). Put theta := max(theta_rel(w) |C_j|, sigma), theta' := theta + r sigma. Then |Z| <= (w+1) e, or a2 is w-FEW on C_j: |C_j \ (S(a2, Lam(a2)) u U_{s} H(a2,s))| <= Gamma' theta' for a set Lam(a2) of <= w + r + C(r,2) radii. Proof. For a real lam: R_lam := (C_j n S(a2,lam)) \ U_s S(s,lam); Z_lam := {a1 in Z : lam notin W(a1)}. Let a1 in Z_lam, x in R_lam \ Ex(a1). x notin S(a2, W(a1)) (|x a2| = lam notin W(a1)); x notin H(s,a2) (else |xs| = lam); if x in H(a1,a2) then |x a1| = lam; if x in H(s,a1) then |x a1| = |xs|; otherwise |x a1| in W(a1). So |x a1|^2 in {v^2 : v in W(a1)} u {lam^2} u {|xs|^2 : s in S}. (3.1) Claim: |R_lam| >= theta => |Z_lam| <= e. Rows R_lam, columns Z_lam (disjoint, C_j n C_l = {}). List W(a1) as v_1(a1), ..., v_w(a1) (repeat entries to pad) and put Bm[x,a1] := prod_{i <= w} (|x a1|^2 - v_i(a1)^2) * (|x a1|^2 - lam^2) * prod_{s in S} (|x a1|^2 - |xs|^2). Each factor |x|^2 + |a1|^2 - (value) - 2 has rank <= D+2; HP: rank Bm <= (D+2)^{k3(w)}. By (3.1) Bm[x,a1] != 0 only for x in Ex(a1): <= delta |C_j| <= (delta/theta_rel(w)) |R_lam| <= |R_lam|/(16 (D+2)^{k3(w)}) per column. GL: R'' x Z'' (|R''| >= floor(|R_lam|/2), |Z''| >= floor(|Z_lam|/2)) with Bm = 0, so [|x a1|^2] is covered there by the k3(w) types v_i(a1)^2 (column functions), lam^2 (constant), |xs|^2 (row functions). LR: R''' x Z''' with one type, |R'''| >= 2^{-L*} |R''|, |Z'''| >= 2^{-L*} |Z''|. Column/constant type: each a1 in Z''' is equidistant from R'''; row type |xs|^2: R''' in H(s, a1) for all a1 in Z'''. OR: min(|R'''|, |Z'''|) <= s'. Since |R_lam| >= theta >= sigma, |R'''| >= 2^{-L*} (sigma - 1)/2 > s'; so |Z'''| <= s', |Z_lam| <= 2^{L+1} s' + 1 <= e. If w+1 distinct radii lam_1..lam_{w+1} have |R_lam_i| >= theta, each a1 in Z misses one of them from W(a1) (|W(a1)| <= w), so Z = U Z_{lam_i} and |Z| <= (w+1) e. Otherwise let Lam_h be the (<= w) radii with |R_lam| >= theta and Lam(a2) := Lam_h u R_{a2} (|R_{a2}| <= r + C(r,2)). If x in C_j, x notin U_s H(a2,s), |x a2| =: lam notin Lam(a2), then |C_j n S(a2,lam)| < theta + r sigma = theta' (|C_j n U_s S(s,lam)| < r sigma). HR_r with c := a2 in W, V := C_j in W \ {a2}, threshold theta' >= sigma bounds these x by Gamma' theta'. []

Step 4 (few w-few centres) [PROVED]

For j notin {l,m}, w in {w1,w2}: the set F of a2 in C_m that are w-few on C_j has |F| <= e. Proof. Rows C_j, columns F (disjoint). Bm[x,a2] := prod_{v in Lam(a2)} (|x a2|^2 - v^2) prod_s (|x a2|^2 - |xs|^2) (Lam(a2) padded to w + r + C(r,2) column functions; it contains |a2 s| and d in D_S): k4(w) factors, rank <= (D+2)^{k4(w)}; nonzero in column a2 only off S(a2,Lam(a2)) u U_s H(a2,s), i.e. <= Gamma' theta' entries, and Gamma' theta' <= Gamma' theta_rel(w) |C_j| + Gamma'(r+1) sigma <= |C_j|/(32 (D+2)^{k4(w)}) + |C_j|/(32 (D+2)^{k4(w)}) by (P4). GL, LR (column functions, constants, row functions |xs|^2), OR exactly as in Step 3, with |R| >= 2^{-L*} floor(|C_j|/2) > s': |F| <= e. [] Corollary 4.1. If Z(a2) in C_l satisfies the hypothesis of Step 3 (for j, w, delta) for every a2 in C_m, then sum_{a2 in

C_m} |Z(a2)| <= e |C_l| + (w+1) e |C_m|. Applications: (a) Z(a2) := {a1 : Good_j(a1,a2)}, w = w1, delta = delta1 (<= theta_rel(w1)/(16 (D+2)^{k3(w1)}) by (P1)); (b) Z(a2) := {a1 : Cov_j(a1,a2)}, w = w2, delta = 6 eps = theta_rel(w2)/(16 (D+2)^{k3(w2)}) (definition of eps).

Step 5 (counting; contradiction) [PROVED; COMPUTER check S5 for (5.1)]

Fix l != m in I. Pi := {(a1,a2) in C_l x C_m : |a1 s| = |a2 s'| for some s, s' in S}; for fixed a2 fewer than r^2 sigma a1 qualify (cap), so |Pi| < r^2 sigma |C_m|. Ty(a1,a2) := D_{S u B} u beta(l) u beta(m) u {|a1 s|} u {|a2 s|} (k5 entries). V_lm := {(a1,a2) notin Pi : |a1 a2| notin Ty(a1,a2)}. (5.1) T is rainbow iff (a1,a2) notin Pi and |a1 a2| notin Ty. [Distances of T: D_{S u B}, |a1 s|, beta(l), |a2 s|, beta(m), |a1a2|; S u B u {a_i} rainbow (F4); beta(l) n beta(m) = {} (F3); |a1 s| notin beta(m), |a2 s| notin beta(l) (F5).] On V_lm: T rainbow; by (F3) at most one k *in I has |a1 a2| in beta(k)*; I0 := I \ {l, m, k*}, |I0| >= 9; for k in I0, beta(k) n D_T = {} (F3; F5 for |a_i s|; k != k)*, so all (k,k') with k != k' in I0 are valid. g := #{k in I0 : Good_k(a1,a2)}. If g <= |I0| - 4, then for each k' in I0 the set J := {k in I0 \ {k'} : not Good_k} has |J| >= 3 and (ii)*{k,k'} for k in J (Step 1), so Cov_k' (Step 2). Hence g + #{k' in I0 : Cov_k'} >= |I0| - 3 >= 6 on V_lm, and summing with Corollary 4.1 over the 10 indices of I \ {l,m}: 6 |V_lm| <= 10 [2e |C_l| + (w1 + w2 + 2) e |C_m|] <= 10 (w1 + w2 + 2) e (|C_l| + |C_m|). Bm0[a1,a2] := prod*{v in Ty(a1,a2)} (|a1a2|^2 - v^2) (k5 factors, rank <= (D+2)^{k5}) vanishes off V_lm u Pi; its density on C_l x C_m is <= (10/6)(w1+w2+2) e (1/|C_l| + 1/|C_m|) + r^2 sigma/|C_l| <= (4(w1+w2+2) e + r^2 sigma)/M <= 5 n^2 sigma / M <= 1/(16 (D+2)^{k5}) [w1+w2+2 = n + (n-2)(n-3) + 2p < n^2; e <= sigma; (P5)]. GL: C_l'' x C_m'' (halves) with Bm0 = 0: [|a1a2|^2] covered by k5 types: constants (D_{S u B}, beta(l), beta(m)), row functions |a1 s|^2, column functions |a2 s|^2. LR: R x C, one type, |R| >= 2^{-L*} *floor(|C_l|/2) > s', |C| >= 2^{-L}* floor(|C_m|/2) > s'. Constant/column type: every a2 in C equidistant from R; row type |a1 s|: R in H(s, a2) for all a2 in C. OR: min(|R|,|C|) <= s'. Contradiction. []

Parameter verification (from M >= M_core, Gamma' >= 1, mu <= sigma, e <= sigma)

Note 1/eps = 96*32 Gamma' (D+2)^{k4(w2)+k3(w2)} <= 32^{10}* Gamma' (D+2)^{k_c}; theta_rel(w1)/(16 (D+2)^{k3(w1)}) = 1/(2^9 Gamma' (D+2)^{k_c}). (P1) delta1 <= 1/(2^9 Gamma' (D+2)^{k_c}): first term <= *62^{10} 4^{n-2} Gamma' (D+2)^{k_c} sigma/M <= 1/(2^{10} Gamma' (D+2)^{k_c}) iff M >= 62^{20}* 4^{n-2} Gamma'^2 (D+2)^{2k_c} sigma: true (2^{20} 4^n n^2 >= *62^{20} 4^{n-2}); second term 3n^2 sigma/M <= 1/(2^{10} Gamma' (D+2)^{k_c}) iff M >= 32^{10}* n^2 Gamma' (D+2)^{k_c} sigma: true. (P2) 2n^2 sigma <= eps M iff M >= *62^{10} n^2 Gamma' (D+2)^{k_c} sigma: true. (P4) M >= 32 (D+2)^{k4(w1)} Gamma' (r+1) sigma: true. (P5) M >= 80 n^2 (D+2)^{k5} sigma: true (k5 <= k_c). Every class has |C_k| >= M > 2^{L+1}*(s'+1) + 1, used for "row side > s'" in Steps 4, 5.

Remarks

- n = 5, r = 0, p = 1: w1 = 7, w2 = 4, k3 = 8/5, k4 = 7/4, k5 = 2: round13/01 (constants differ slightly).
- r = 0: Hyp, Sm, Pi are empty and R_{a2} = {}: only column functions and constants occur. The general r case adds row types |xs|^2 (bisectors H(s,a_i)) and small classes of s (cap sigma, costs absorbed in delta1, eps, Step 5).
- Validity is a CONSTANT-type condition (|a1a2| in beta(k*)) plus the Step-5 type list; this is what the family conditions (F3), (F5) buy, and why more than two auxiliary points are never used.

Part 3

03. Slicing lemma (levels r < n-5), E1_r, threshold constants

STATUS: Slicing Lemma PROVED; Theorem 1.4 (E1_r) PROVED given HR_r and the Core Lemma (02). Elementary inclusion of step (d) COMPUTER-checked (code/check_core.py, check SL). Uses HR_r, GL, LR, OR, HP, Konig, Caro-Wei.

Setting as in 02: level-r configuration (S, W, sigma), sigma >= sigma_min, HR_r with constant Gamma'; rainbow families (F1)-(F5) as defined in 02. e := 2^{L*+1}(s'+1) <= sigma.

Slicing Lemma

Let 0 <= r <= n-6, F = (B, (C_k, beta(k))*{k in I}) a rainbow family at level r with 1 <= p := |B| <= n-5-r, |I| >= 6K (K >= 1). Put k_sl := 3K - 1 + C(r+p,2) + 2p + 2r, theta_sl := 1/(16 Gamma' (D+2)^{k_sl}), M_sl := 2n(n+K) sigma / theta_sl. If M(F) >= M_sl, there are c in U_k C_k and a rainbow family F' = (B u {c}, (C'_k, beta'(k))*{k in I'}) at level r with I' in I, |I'| = K, C'_k in C_k and M(F') >= theta_sl M(F)/2.

Proof

Restrict to |I| = 6K; M := M(F). Fix pi in I. For c in C_pi, S u B u {c} is rainbow (F4) and Phi(c) := D_{S u B} u beta(pi) u {|cs| : s in S} = D_{S u B u {c}} (|c b_i| = beta_i(pi)); |Phi(c)| <= C(r+p,2)+p+r; R_c = D_S u {|cs|} is contained in Phi(c). Graph G_c (bipartite; left I \ {pi}, right the reals): k ~ lam iff lam notin Phi(c) u beta(k) and |C_k n S(c,lam)| >= theta_sl |C_k|. (Each k has <= 1/theta_sl neighbours; G_c is finite.)

(a) Few bad centres: Z_bad := {c in C_pi : G_c has no matching of size 3K} has |Z_bad| <= 2e. For c in Z_bad, Konig's theorem gives a vertex cover A(c) u Lam0(c) with |A(c)| + |Lam0(c)| <= 3K - 1. Call c k-FEW for every k in I \ ({pi} u A(c)). Let c be k-few and x in C_k with x notin U_s H(c,s) and lam := |xc| notin Lam0(c) u Phi(c) u beta(k). Then (k,lam) is not an edge (k notin A(c), lam notin Lam0(c)), and lam notin Phi(c) u beta(k), so |C_k n S(c,lam)| < theta_sl |C_k|; also lam notin R_c. HR_r (c in W, V := C_k in W \ {c}, threshold theta := theta_sl |C_k| >= theta_sl M >= sigma) bounds such x by Gamma' theta_sl |C_k| = |C_k|/(16 (D+2)^{k_sl}). Hence Bm_k[x,c] := prod_{v in Lam0(c) u Phi(c) u beta(k)} (|xc|^2 - v^2) * prod_{s in S} (|xc|^2 - |xs|^2) (Lam0(c) padded to 3K-1 entries; (3K-1) + |Phi| + p + r <= k_sl factors; rank <= (D+2)^{k_sl} by HP) has at most |C_k|/(16 (D+2)^{k_sl}) nonzero entries in each column c that is k-few. Counting: sum_{c in Z_bad} #{k in I \ {pi} : c is k-few} >= |Z_bad| ((6K - 1) - (3K - 1)) = 3K |Z_bad|, so some k has Z_k := {c in Z_bad : c k-few} with |Z_k| >= 3K |Z_bad| / (6K - 1) >= |Z_bad|/2. Rows C_k, columns Z_k (disjoint). GL: C_k'' x Z_k'' (halves) with Bm_k = 0, so [|xc|^2] is covered there by k_sl types: column functions (entries of Lam0(c), |cs|^2), constants (D_{S u B}, beta(pi), beta(k)), row functions |xs|^2. LR: single-type R x C, |R| >= 2^{-L*} *floor(|C_k|/2) > s'. Column/constant type: every c in C equidistant from R; row type: R in H(s,c) for all c in C. OR: |C| <= s'. So |Z_k| <= 2^{L+1}* s' + 1 <= e and |Z_bad| <= 2e.

(b) Choice of c. |C_pi| >= M >= M_sl > 2 sigma >= 2e, so some c in C_pi \ Z_bad exists. Fix it and a matching {(k, lam_k) : k in I_1} of G_c, |I_1| = 3K (the lam_k are distinct).

(c) Label conflicts. By (F3) each lam_k equals beta_i(k') for at most one pair (i,k'). The digraph on I_1 with arcs k -> k' (k' != k, lam_k in beta(k')) has out-degrees <= 1, hence <= 3K edges and average degree <= 2; Caro-Wei gives an independent set of size >= 3K/3 = K. Let I' be K of its elements.

(d) New family. B' := B u {c}; for k in I': beta'(k) := (beta(k), lam_k) and C'*k := (C_k n S(c, lam_k)) \ U*{s in S} S(s, {|cs'| : s' in S} u {lam_k' : k' in I'}). |C'*k*| >= theta_sl |C_k| - r(r+K) sigma >= theta_sl M - n(n+K) sigma >= theta_sl M / 2 (theta_sl M >= 2n(n+K) sigma). (F1) S u B u {c} rainbow (F4 for c). (F2) |x b_i| = beta_i(k), |xc| = lam_k on C'_k. (F3) Old values pairwise distinct (F3); lam_k pairwise distinct (matching); lam_k notin beta(k) (edge) and lam_k notin beta(k') for k' in I' \ {k} (independence). None lies in D{S u B u {c}} = Phi(c): lam_k notin Phi(c) (edge); beta_i(k) notin D_{S u B} (F3), beta_i(k) != beta_j(pi) (F3, k != pi), beta_i(k) != |cs| (F5 for c). (F4) For x in C'*k the distances of S u B u {c,x} are those of S u B u {x} (rainbow), those of S u B u {c} (rainbow), and |xc| = lam_k. Cross coincidences: |cs| = |xs'| (removed); beta_j(pi) = |xs| (F5); |cs| = beta_i(k) (F5 for c); beta_j(pi) = beta_i(k) (F3); lam_k in D*{S u B} u beta(pi) u {|cs|} (lam_k notin Phi(c)); lam_k in beta(k) (edge); lam_k = |xs| (removed). [COMPUTER check SL.] (F5) |xs| notin beta(I') (old F5) and |xs| != lam_k' (removed). So F' is a rainbow family with p+1 centres, |I'| = K, M(F') >= theta_sl M/2. []

Theorem 1.4 (E1_r) [PROVED given HR_r]

Fix 0 <= r <= n-5. For p = 1..n-4-r put K^(p) := 12 * 6^{n-4-r-p} (so K^(1) = K_r = 12*6^{n-5-r}, K^(n-4-r) = 12). For p <= n-5-r: k_sl(p) := 3K^(p+1) - 1 + C(r+p,2) + 2p + 2r, theta_sl(p) := 1/(16 Gamma' (D+2)^{k_sl(p)}), M_sl(p) := 2n(n + K^(p+1)) sigma/theta_sl(p). Define m_{n-4-r} := M_core (02) and, downwards, m_p := max(M_sl(p), 2 m_{p+1}/theta_sl(p)) (p = n-5-r, ..., 1). Then ratio_r := 2 m_1/sigma + 2n(n + K_r) (ratio) (a number depending only on n, D, Gamma'), and every b in W has |Big_b(ratio_r sigma)| < K_r. Proof. tau := ratio_r sigma. Suppose b in W has K_r radii rho_k (k in I) in Big_b(tau). Initial family: B := {b}, beta(k) := (rho_k), C_k := (W n S(b,rho_k)) \ U_{s in S} S(s, {|bs'| : s'} u {rho_k' : k' in I}). |C_k| >= tau - r(r + K_r) sigma >= tau/2 >= m_1. (F1) S u {b} rainbow (b in W). (F2) clear. (F3) rho_k distinct, notin R_b = D_{S u {b}} (definition of Big_b). (F4) x in C_k: S u {x} rainbow, |xb| = rho_k notin R_b, |xs| != |bs'| and |xs| != rho_k (removed), |xs| != |xb| likewise: S u {b,x} rainbow. (F5) removed. For p = 1, ..., n-5-r apply the Slicing Lemma with K := K^(p+1) (|I| = K^(p) = 6K, M >= m_p >= M_sl(p)): a family with p+1 centres, K^(p+1) classes and M >= theta_sl(p) m_p / 2 >= m_{p+1}. At p = n-4-r: 12 classes, M >= M_core, contradicting the Core Lemma. (r = n-5: no slicing, ratio_{n-5} = 2 M_core/sigma + 2n(n+12).) []

Constants and sizes

k := *max(C(n-2,2) + 2(n-2), k'(R+), max_r k_c(r), max_{r,p} k_sl(r,p)) = O(6^n) [k_sl <= 3126^{n-6} + n^2]; L* := L_{k*}*(D+2) = O(k*^3 sqrt(D+2) log(D+2)); e := 2^{L+*1}*(s'+1). *sigma_abs := max(mu, e, 2^{L*+1}(s'+1) + 2)* (used: mu <= sigma in delta1; e <= sigma in Steps 3-5 and (b); row sides > s'). Size of ratio_r: M_core/sigma = 2^{20} 4^n n^2 Gamma'^2 (D+2)^{2k_c}, each slicing round multiplies by 2/theta_sl = 32 Gamma' (D+2)^{k_sl}: ratio_r <= C_n Gamma'^{n-3-r} (D+2)^{2k_c + sum_p k_sl(p)} = (D+2)^{O_n(1)} when Gamma' = (D+2)^{O_n(1)}. Exceptional choices in the slicing step: (i) c in Z_bad (<= 2e points, avoided); (ii) small classes of s through c or with radius lam_k' (removed, r(r+K) sigma points per class); (iii) label conflicts lam_k in beta(k') (independent set, factor 3); (iv) radii in Phi(c) u beta(k) (never edges; they enter the few-cover as constants or column functions); (v) bisectors H(c,s) (row types).

Part 4

04. Test case n = 6, written out (instance of 01-03)

STATUS: PROVED (candidate), as an instance of 01-03; all loci lists below were COMPUTER-checked for n = 6 (r = 1, p = 1 and r = 0, p = 2) in code/check_core.py. Conditional on the cited in-project results (LR, G1, R+, Lemma H step of Theorem R) exactly as the general proof.

n = 6: |T| = n-2 = 4. Levels j = 4 (base), 3, 2 (= n-4): Theorem R+ (01, Thm 1.3), k' = C(4,2) + 12 = 18. E1 needed at levels r = 1 (= n-5) and r = 0.

Level r = 1 (S = {s}, one spread point; B = {b}; core lemma only)

Configuration (S = {s}, W, sigma): every class of s meets W in < sigma points; s u {x} rainbow means x != s only. HR_1 (from Claim(2) = R+): for c in W, V in W{c}, theta >= sigma: #{x in V : |xc| != |cs|, x notin H(c,s), |V n S(c,|xc|)| < theta} <= (Gamma_2 + 1) theta. E1_1: every b in W has at most 11 radii lam != |bs| with |W n S(b,lam)| >= ratio_1 sigma. Proof = Core Lemma with r = 1, p = 1, B = {b}, beta(k) = rho_k. Classes cleaned: C_k := W n S(b,rho_k) minus S(s, {|bs|} u {rho_k'}) (< 13 sigma points). T = {s, b, a1, a2}; D_T = {|sb|, |s a1|, |s a2|, rho_l, rho_m, |a1a2|}. Step 0: x in C_k non-generic => x in S(a_i, D_T u {rho_k}) u H(a1,a2) u H(s,a1) u H(s,a2) u S(s, {|s a1|, |s a2|, |a1a2|}). Step 1: L_x = {|xs|, rho_k, |x a1|, |x a2|}; Q' = {w : rho_k notin {|w a1|, |w a2|}} (|ws| != rho_k by cleaning); Good_k: C_k in S(a_i, W) u H(a1,a2) u H(s,a1) u H(s,a2) u Ex, |W| <= w1 = 4 + 6 + 1 = 11; Cov_k': W = D_T u {rho_k'}, w2 = 7. Step 3 (a2 fixed): rows R_lam = C_j n S(a2,lam) \ S(s,lam); types for |x a1|^2: <= 11 column functions, lam^2, and the ROW function |xs|^2 (bisector H(s,a1)); k3(w1) = 13. Few branch: HR_1 at a2 leaves C_j on <= 11 + 1 spheres about a2 (heavy radii and |a2 s|) and on H(a2,s) (row type |xs|^2): k4(w1) = 13. Step 5: Pi = {|a1 s| = |a2 s|}; Ty = {|sb|, rho_l, rho_m, |a1 s|, |a2 s|} (k5 = 5): constants, one row, one column type; OR for the row type |a1 a2| = |a1 s|: R in H(s,a2) for all a2 in C. k_c = 26, M_core = 2^{20} 4^6 36 Gamma'^2 (D+2)^{52} sigma, ratio_1 = 2 M_core/sigma + 216, K_1 = 12. Claim(1): Lemma 1.2 with K = 11: Gamma_1 = C(D+14,13)^2 (2 ratio_1 Gamma_2 + 3).

Level r = 0 (S = {}; one slicing round, then the core lemma with B = {b, c})

HR_0 (from Claim(1)): for c in W, V in W \ {c}, theta >= sigma: #{x in V : |V n S(c,|xc|)| < theta} <= Gamma_1 theta (no bisector exception at r = 0). E1_0: every b in W has at most 71 radii with |W n S(b,lam)| >= ratio_0 sigma. Proof. Suppose b has 72 such radii; C_k := W n S(b,rho_k) (no cleaning at r = 0), beta(k) = (rho_k), K^(1) = 72. Slicing (p = 1 -> 2, K = 12): fix pi; for c in C_pi, Phi(c) = {rho_pi} (= |bc|). G_c: k ~ lam iff lam notin {rho_pi, rho_k} and |C_k n S(c,lam)| >= theta_sl |C_k|, theta_sl = 1/(16 Gamma_1 (D+2)^{37}) (k_sl = 3*12 - 1 + 0 + 2 + 0 = 37). *If G_c has no matching of size 36, Konig gives A(c), Lam0(c) (|A| + |Lam0| <= 35), and for k notin A(c) u {pi}, HR_0 puts all but |C_k|/(16 (D+2)^{37}) points of C_k on the spheres about c with radii Lam0(c) u {rho_pi, rho_k}; |Z_bad| <= 2e by averaging (some k is few for >= half of Z_bad), GL, RL (column functions and constants only), OR. Take c notin Z_bad, a 36-matching (k, lam_k), and an independent set of 12 indices in the conflict digraph (lam_k = rho_k'). New family: B = {b, c}, beta(k) = (rho_k, lam_k), C'k = C_k n S(c, lam_k), |C'_k| >= theta_sl |C_k|. (F3): the 24 values rho_k, lam_k are distinct and avoid D{b,c} = {rho_pi}. Core lemma (r = 0, p = 2): T = {b, c, a1, a2}, D_T = {rho_pi, rho_l, lam_l, rho_m, lam_m, |a1 a2|}.*

*Step 0: x in C'_k non-generic => x in S(a_i, D_T u {rho_k, lam_k}) u H(a1,a2) (no other loci at r = 0). Good: |W| <= w1 = 4 + 6 + 2 = 12; Cov: w2 = 8; k3(w1) = 13, k4(w1) = 12, k_c = 25. Step 2 uses that the radius PAIRS {rho_k, lam_k} are pairwise disjoint across k (F3). Step 5: Ty = {rho_pi, rho_l, lam_l, rho_m, lam_m} (k5 = 5, all CONSTANTS): GL + RL + OR. K_0 = 72; ratio_0 = 2 m_1/sigma + 26*78* with m_2 = M_core(r=0) = 2^{20} 4^6 36 Gamma_1^2 (D+2)^{50} sigma, m_1 = max(M_sl, 2 m_2/theta_sl) = 32 Gamma_1 (D+2)^{37} m_2. Claim(0): Lemma 1.2 with K = 71: Gamma_0 = C(D+72,71)^2 (2 ratio_0 Gamma_1 + 1).

Conclusion for n = 6

|X| <= Gamma_0 sigma_min with Gamma_0 = (D+2)^{O(1)} and sigma_min = 2^{O(L_37(D+2))} (s'+1) (k* = 37 here). Exponent bookkeeping (HEURISTIC, constants C0 ignored, P <= (D+2)^{2e}, Q ~ (D+2)^{216}): log_{D+2} of Gamma_3, Gamma_2, Gamma_1, Gamma_0 ~ 238, 468, 1.5e3, 6.2e3. Hence s'_6(D) <= 2^{C sqrt(D) log(D+2)} (s'_6(floor(D/2)) + 1) and f_d(6) = 2^{O(sqrt d log d)}.

Part 5

05. Adversarial self-check of 01-04 (no independent referee; everything re-derived by hand)

Verdict: no gap found in the new parts (Lemma 1.1, Lemma 1.2, Core Lemma, Slicing Lemma, E1_r, tower). The result is CONDITIONAL on in-project results that have had AI referee passes only (LR, G1, Theorem R, Theorem R+), and it re-derives (in generalised form) the round-13 n = 5 argument, which is itself still under separate refereeing: any flaw found there propagates here.

A. Dependency audit

used	where	source	status there
G1 (P x Q form)	Core Step 1	round12/01 Tools; round11/02 + LR-4	PROVED; referee12 "correct"
Theorem LR (row, column, constant types)	Core Steps 3-5, Slicing (a)	round11/01	PROVED in-project; referee pass recorded in round11/00_index; NEW theorem, no external check
GL, HP, OR, DIM	throughout	round9/01, round11/n4_full_proof Sec. 1	CITED (GL: Gavinsky-Lovett, HMST Lemma 4.7) / PROVED
Lemma H, Caro-Wei	Lemma 1.2	round8; classical	PROVED / CITED
Konig's theorem	Slicing (a)	classical	CITED
Theorem R+ (levels >= n-4)	Thm 1.3	round12/03 + REFEREE_R.md F1-F3	PROVED (AI referee "correct with fixes"); single-cap restatement here NOT separately refereed
halving recursion	Thm 1.6	round11/n4_full_proof Sec. 9	PROVED

used	where	source	status there
Not used: Lemma S2 outside R+, heaviness G2, Props 4.6-4.8, the round-12 "level switch", absolute E1.			

B. Step-by-step attack

1. Lemma 1.1 (HR_r). The list of non-rainbow events of S u {c,x} is complete (COMPUTER "HR": equivalence, 0 violations). The configuration (S u {c}, V_1, theta) satisfies the cap for s (V_1 in W) and for c (V_1 in V). Claim(r+1) is a statement about ALL level-(r+1) configurations with cap >= sigma_min, so a free threshold theta is allowed: this is item 1 of round13/03 Section 4 ("towers uniform in the base"), settled by the configuration formalism (every constant is a ratio; Gamma_j does not depend on sigma).
2. Bisector exception (round13/03 item 3). Points of V on H(c,s) are excluded from HR and re-enter every application as the ROW type |xc|^2 = |xs|^2 (Core Step 3: H(s,a1) in (3.1); Step 4: H(a2,s); Slicing (a): H(c,s)). Points on H(s,a2) inside a row slice R_lam lie on S(s,lam) and are removed from R_lam (< r sigma). Each single-type rectangle of row type has R in H(s,col) for every column: OR applies (p := s fixed). Checked in all three places.
3. Spread-point coincidences (item 2). Step 0 lists every coincidence of D_T u L_x (COMPUTER "G": 0 violations; 63,070 instances, 33,899 non-generic, n = 6, 7, all (r,p); totals in code/check_core.log). Coincidences through s are (a) impossible by F4/F5, (b) on <= r(2r+1) spheres of s (cap), (c) bisectors H(s,a_i) (row types). Step 1(i) adds r(r+2) spheres (COMPUTER "S1"). Their total relative cost 3n^2 sigma/M is in delta1, and in (ii) it is absorbed by 2n^2 sigma <= eps M. T rainbow is characterised exactly (5.1; COMPUTER "S5"): pair-small events Pi (density < r^2 sigma/M) and the k5 LR types.
4. Validity. With (F3),(F5), the only k-dependent invalidity is |a1a2| in beta(k*) for one k* (F3: all values distinct). This is what makes q >= 3 harmless: we never use more than two auxiliary points; the remaining n-4-r points of T are centres with CONSTANT labels on each class (the "validity problem" of round13/03 0(c) does not arise).
5. Step 2 with multi-centre labels: needs the radius SETS beta(k) pairwise disjoint (F3), not just one radius each. OK.
6. Two-centre lemma. Column functions: W(a1) depends on a1 (L_w0 with w0 = w0(a1,a2), D_T contains |a1 s|, |a1 a2|), fixed a2: fine. Constant lam. Rows |xs|^2. k3(w) types. Per-column nonzeros <= delta|C_j| <= |R_lam|/(16 (D+2)^{k3}) because |R_lam| >= theta >= theta_rel |C_j|. Pigeonhole over w+1 heavy radii: correct. Few branch: R_{a2} included in Lam(a2) (HR does not see S(a2,R_{a2})).
7. Step 4 and Corollary 4.1: unchanged from n = 5 except the type count k4(w) = w + 2r + C(r,2).
8. Step 5: 10 indices, (10/6) factor, density bound 5n^2 sigma/M; LR with constants, rows |a1 s|^2, columns |a2 s|^2; every type gives OR. Both sides > s'. OK.
9. Slicing (a): HR applies (c in W, C_k in W \ {c}, theta >= sigma). Points of C_k on S(c, beta(k)) are on H(c,b_i); they are a CONSTANT-radius sphere for the fixed row class k: fine. Averaging needs |I| - 1 - (3K-1) >= (|I|-1)/2, true for |I| = 6K. GL per-column density bound implies the global one.
10. Slicing (c)-(d): conflict digraph out-degree <= 1 because all old labels are distinct (F3); new labels avoid Phi(c) = D_{S u B u {c}} by the edge rule; c's own labels |cs|, beta(pi) are kept away from old labels by F3/F5; cleaning of S(s,|cs'|) and S(s,lam_k') restores F4/F5 (COMPUTER "SL": 0 violations).
11. Lemma 1.2: identical to Theorem R's step (referee12 item 1 "correct") with Typ replaced by the relative Big_b.

12. Thresholds. Order of definition: Gamma_{n-2..n-4} -> (Gamma'_r -> ratio_r -> Gamma_r) for r = n-5..0 -> sigma_min. ratio_r uses only Gamma'_r, n, D; sigma_min is a max of absolute quantities (mu, e, e_A, tau_min^{R+}, *24^{n-2} mu + 4, 2^{L*+1}(s'+1) + 2) none of which involves sigma_min or a Gamma: no circularity. Uses of "absolute <= sigma": mu (delta1), e (Steps 3-5, slicing (b)), s' (row sides), e_A (R+ recursion). Every constraint (P1)-(P5), M_sl, initial cleaning tau >= 2n(n+K_r) sigma is implied by the definitions (02, 03).
13. Growth. g_r := log_{D+2} Gamma_r. ratio_r <= C_n Gamma'_r^{n-3-r} (D+2)^{O(n k)} *and Gamma_r ~ P ratio_r Gamma_{r+1}, so g_r <= (n-2-r) g_{r+1} + O(n k)*: g_0 = n^{O(n)} 6^{O(n)}, a constant for fixed n. Gamma_0 = (D+2)^{O_n(1)}.

C. Things that would break the proof (none found)

- If HR needed Claim at the SAME level (threshold escalation): avoided, HR_r uses Claim(r+1) only.
- If some coincidence were neither small, nor LR-type, nor excluded by F3-F5: none found by hand; none by computer.
- If the Konig alternative did not give a covering statement: it does (vertex cover + HR => all but Gamma' theta_sl |C_k| points of C_k on <= k_sl spheres/bisectors), so only COVERING-mode GL is used (density <= 1/(16 rank)); the log-rank density barrier (round12/02 (C5)) is never met.
- Consistency: (C2)-type example X = Y u {o} (Y rainbow-(n-1)-free): E1_0 says o has <= K_0 - 1 classes of size

  = tau_0 in X. This is consistent iff f_D(n-1) < K_0 tau_0 or o cannot be placed; nothing known contradicts it (no superpolynomial lower bounds on f_D(n-1) are known to us). Hierarchy examples (round9/02): fine (K >= 2 for n >= 6).

D. Residual risks (honest)

1. LR is a new rank-rectangle theorem (would improve published Singer-Sudan bounds for parallel partitions); if LR fails, G1, R+ and every step here fail.
2. The R+ restatement with a single cap (01 Thm 1.3) is a translation checked here, not by the round-12 referee.
3. The n = 5 core argument (round13) is under separate refereeing; this file generalises it.
4. Constants are not optimised and were checked by hand only (not by computer algebra).

Appendix A. Theorem LR (label rounding) — round 11

Source file: e1088/round11/01_label_rounding.md

01. Label rounding: low-rank matrices whose entries are row labels or column labels

STATUS: PROVED (new this round; self-checked line by line; independent referee pass recorded in 00_index.md). This settles Hypothesis LRQ of round10/06 with L(r) = O(sqrt(r) log r), hence (BIP) as well.

Theorem LR

Let M in R^{m x n} have rank <= r (r >= 1). Let T = {tau_1, ..., tau_k} be "type functions" on [m] x [n], each either a ROW function (tau(x,w) = a(x)) or a COLUMN function (tau(x,w) = b(w)); constants count as both. Suppose every entry is covered: for all (x,w) there is tau in T with M[x,w] = tau(x,w). Then there are R in [m], C in [n] and ONE type tau *in T with* |R| >= 2^{-L} m, |C| >= 2^{-L} n, M[x,w] = tau(x,w) on R x C, L = L_k(r) <= C0 k^3 sqrt(r+2) log2(r+2) (C0 an absolute constant).

Special cases: all types column functions = RQ-list (round 9); all types constants = RQ; three column values plus two row labels = Hypothesis LRQ (round10/06), for which the single-type conclusion is stronger than needed.

Why margins are not lost (the one new idea)

Rounding (Lemma R below) needs a sign pattern with bounded aspect ratio. Earlier rounds normalised only by GLOBAL or per-COLUMN constants, and labels f(x) accumulating at another value (e.g. f(x) = kappa + 2^{-j}) destroy every such margin (round10/03, "accumulation obstruction"). But every matrix N = diag(alpha) (M - phi 1^T - 1 theta^T) diag(beta) (alpha, beta > 0) has rank <= r + 2 and the same type pattern. Gaps between type values are differences of row and column functions, and on a sub-rectangle of CONSTANT relative size each such gap is comparable to a row function, a column function or a constant (Lemma B). So the largest gap can be normalised to size ~1 by a row OR column rescaling, and its midpoint is of the form phi(x) + theta(w). The accumulation obstruction disappears.

Tools (all PROVED in round9/01, re-checked this round)

(R) Lemma R (margin rounding; Rothvoss arXiv:1409.6366 Sec. 3 made quantitative). N in R^{m x n}, rank s, 1/2 <= |N_xw| <= K for all entries, Q+ = {N > 0} with |Q+| >= delta m n, 0 < delta <= 1/2. Then there is X' x Y' with |X'| >= (delta/2) 4^{-T} m, |Y'| >= (delta/2) 4^{-T} n and at most delta |X'||Y'| entries of Q- = {N < 0} in it, T = ceil(pi K sqrt(s) ln(2/delta^2)). [John position: |u_x| <= s^{1/4}, |v_w| <= K s^{1/4}; normalised margin >= 1/(2K sqrt s); Sheppard's formula; T independent Gaussian cuts.] (GL) Cleaning lemma (Gavinsky-Lovett; HMST Lemma 4.7). rank B <= R' and at most |A||B|/(16R') nonzero entries => B vanishes on some A'' x B'' with |A''| >= |A|/2, |B''| >= |B|/2. (HP) rank(A o B) <= rank A * rank B (Hadamard product).

Terminology. A REFINEMENT of a rectangle R0 x C0 with cost (c1, c2) is a sub-rectangle R x C with |R| >= |R0|/c1, |C| >= |C0|/c2. All properties below are inherited by refinements.

Lemma A (separating a row function from a column function). [PROVED]

For a row function a and a column function b on R0 x C0 there is a refinement of cost (4, 2) on which either (i) a is constant, or (ii) there is theta with a(x) > theta >= b(w) for all (x,w) [or a(x) < theta <= b(w)]. Proof. theta := a median of a on R0. If |{a = theta}| >= |R0|/4, take R = {a = theta}, C = C0: (i). Otherwise |{a > theta}|, |{a < theta}| >= |R0|/4. One of {b <= theta}, {b >= theta} has >= |C0|/2 columns; pair it with {a > theta}, resp. {a < theta}: (ii). []

Lemma B (trichotomy of scales). [PROVED]

Let D > 0 be a row function and E >= 0 a column function on R0 x C0. There is a refinement of cost (4, 4) on which (a) D >= 2E everywhere, or (b) E >= 2D everywhere, or (c) for some s > 0, D, E in (s/4, 4s) everywhere. Proof. s := a median of E on C0. If s = 0: {E = 0} has >= |C0|/2 columns and D > 0 = 2E: (a). Let s > 0. If |{D >= 2s}| >= |R0|/4: {D >= 2s} x {E <= s}: (a). Else if |{D <= s/2}| >= |R0|/4: {D <= s/2} x {E >= s}: (b). Else R1 = {s/2 < D < 2s} has >= |R0|/2 rows. If |{E <= s/4}| >= |C0|/4: R1 x {E <= s/4}: (a). Else if |{E >= 4s}| >= |C0|/4: R1 x {E >= 4s}: (b). Else R1 x {s/4 < E < 4s}, |C| >= |C0|/2: (c). []

Call a function Delta on R x C c-COMPARABLE to Q if Q <= |Delta| <= c Q there, where Q is a BASIC quantity: a positive row function, a positive column function, or a positive constant.

Lemma NF (normal form). [PROVED]

Let T cover all entries of M on R0 x C0, |T| = k. There is a refinement of cost (2^{c k^2}, 2^{c k^2}) (c absolute) and a subset T' of T, k' = |T'| >= 1, still covering all entries, such that on R x C: (NF1) the k' types of T' are strictly ordered, tau_(1)(x,w) < ... < tau_(k')(x,w), with the SAME order for all (x,w); (NF2) if k' >= 2, every gap g_j := tau_(j+1) - tau_(j) is 16-comparable to a basic quantity Q_j; (NF3) if k' >= 2, there is j *with Q_{j}* >= Q_j / 16 for all j. Proof. Treat the C(k,2) pairs (tau, tau') one after another, refining each time: - both row functions: split rows by the sign of tau - tau' (3 parts, cost (3,1)); if the part "=" is kept, tau and tau' coincide on the rectangle: delete tau' from T (coverage is kept). Otherwise |tau - tau'| is a positive row function: 1-comparable. Two column functions: symmetric. - a row function a and a column function b: Lemma A. In case (ii), a - b = D + E with D = |a - theta| > 0, E = |theta - b| >= 0 (or the mirror image); Lemma B gives (a) D <= |a-b| <= 1.5 D, (b) E <= |a-b| <= 1.5 E (E > 0 as D > 0 and E >= 2D), (c) s/2 < |a-b| < 8s = 16 (s/2). In case (i) a is constant and is treated as a column function from now on: split columns by the sign of a - b (cost (1,3)), deleting b if "=" is kept. After all pairs, every pairwise difference of surviving types has constant sign (giving NF1, the order being total) and is 16-comparable to a basic quantity (NF2). For NF3: split rows by which row-type Q_j is largest (<= k parts) and columns likewise; call the winners R*(x), C*(w); let s *be the largest constant Q_j. Lemma B on (R*, C*), then split rows (case a) or columns (case b) by comparison with s*; in case (c) compare the constants s and s. *In every case one Q_{j}* is >= 1/16 of every Q_j (case (c), s < *s: R* >= s/4, C <= *4s* <= *16 R*, s < *s* <= *4 R*). Degenerate cases: if there is no row-type (column-type) Q_j, skip R *(C)* and only compare the other winner with s*; if there is no constant Q_j put s* := 0. Total cost 2^{O(k^2)} per side. []

Proof of Theorem LR

Induction on the number of types. If k = 1 there is nothing to do. Bring Ro x Co = [m] x [n] into normal form (Lemma NF); if k' = 1 we are done. Otherwise, by NF2-NF3, g_{j} >= Q_{j} >= max_j Q_j / 16 >= max_j g_j / 256, and, since g_j <= 16 Q_j <= 256 Q_{j}, the spread S := tau_(k') - tau_(1) = sum_j g_j <= 256 (k'-1) Q_{j}. Put z := (tau_(j) + tau_(j+1)) / 2 (= phi(x) + theta(w): each tau is a row or column function), sigma := Q_{j} (= alpha(x)^{-1} beta(w)^{-1}: a row fn, a column fn or a constant), N := (M - z) / sigma = diag(alpha) (M - phi 1^T - 1 theta^T) diag(beta), rank N <= r + 2. An entry of type tau_(i) with i <= j has N <= -g_{j}/(2 sigma) <= -1/2 and N >= -S/sigma >= -256 (k'-1); with i > j it has 1/2 <= N <= 256 (k'-1). (By NF1 each entry has exactly one type in T'.) So 1/2 <= |N| <= K with K = 256 (k-1). Negate N if necessary so that Q+ is the majority side; let S_min be the types on the minority side, S_maj the others. Put delta := 1/(16 (r+1)^{k-1}). (If the minority side has no entries, skip Lemma R and (GL) in this round. All "halves" are rounded down; every rectangle produced stays nonempty.) Lemma R gives X' x Y' (relative sizes >= (delta/2) 4^{-T} each, T = ceil(pi K sqrt(r+2) ln(2/delta^2))) with at most a delta-fraction of minority entries. The minority entries are EXACTLY the support of H := o_{tau in S_maj} (M - tau) [(M - tau) = M - a 1^T or M - 1 b^T, rank <= r+1], because by NF1 a minority-type value differs from every majority-type value at the same (x,w). By (HP) rank H <= (r+1)^{|S_maj|} <= (r+1)^{k-1}, so (GL) gives a half x half sub-rectangle with H = 0: there every entry has a type in S_maj, |S_maj| <= k' - 1 < k. Apply the induction hypothesis to the k - 1 or fewer types S_maj. Loss per round: O(k^2) bits (Lemma NF) + 1 + log2(2/delta) + 2T = O(k^2 sqrt(r+2) log(r+2)) bits; at most k - 1 rounds; total L = O(k^3 sqrt(r+2) log(r+2)). []

Corollaries

LR-1 (Hypothesis LRQ of round10/06 holds). [PROVED] Rank-r M with M[x,w] in V_w u {f1(x), f2(x)}, |V_w| <= 3: pad V_w to three column functions c1(w) < c2(w) < c3(w) (a padded value need not occur), add f1, f2: k = 5. LR gives a rectangle of relative size 2^{-O(sqrt r log r)} with a single type: either every column is constant on R, or M = f_i on R x C. So LRQ(L) holds with L(r) = O(sqrt(r) log r). LR-2 ((BIP) holds). [PROVED] By round10/06 Remark 6.2 / Step 1: mu_BIP(D) = 3 * 2^{L(D+2)} (s'(floor(D/2)) + 1). LR-3 (round 9 Label-RQ, 3 global values + a row label + a column label). [PROVED] k = 5, L = O(sqrt r log r). LR-4 (general n). [PROVED] Lemma 5.1 of round10/05: for T-generic, label-disjoint x, w the value |xw|^2 lies in {d^2 : d in D_T} (constants), {|w t_i|^2} (column functions), {|x t_i|^2} (row functions): k = C(n-2,2)+2(n-2) types. LR gives a single-type rectangle; every type gives an orthogonal split (constant or |w t_i|: w equidistant from R; |x t_i|: R in H(t_i, w)). So the bipartite step (Theorem 5.4 (1) of round10/05) is unconditional with loss 2^{O_n(sqrt D log D)}.

Consistency checks (HEURISTIC / literature)

- Novelty flag (referee): Singer-Sudan (arXiv:2101.09592) state that the best known bound for PARALLEL 3-partitioned configurations is 2^{-O~(d)}. RQ-list (round 9; = LR with column types only) gives 2^{-O(sqrt d log d)} for parallel k-partitions, every fixed k. If correct this already improves the published state of the art; we found no later paper doing so, but the search was brief. Both statements need external checking before any use outside this project.
- Boolean rank-r matrices: LR returns 2^{-O(sqrt r log r)} monochromatic rectangles = Lovett/Rothvoss. LR does not beat the log-rank barrier (polylog would be needed for that), so no conflict with the round10/02 calibration.
- Singer-Sudan Conj. 2.7 (non-parallel k-partitions) is NOT proved here. LR covers only partitions in which the equation of each hyperplane of a block reads = tau(x,w) with tau a row or a column function; in the lifted space (u_x, a_1(x), a_2(x), ...) all normals of a block agree modulo a FIXED subspace. That structure is exactly what the 4-sets {o,a,x,w} produce (Lemma 2.1 of round10/02 realises it with normal w - a), and it is what makes the margins after rescaling. General non-

parallel partitions (two unrelated low-rank matrices A, A' with A ∘ A' = 0) have no such rescaling, and Singer-Sudan Thm 1.8 (density 1/sqrt d) is not contradicted: LR uses the covering, not density.

- The constants are huge (K = 256(k-1), C0 ~ 10^6); only the shape 2^{-O(sqrt r log r)} matters here.

Appendix B. Lemma R and the cleaning lemma GL — round 9

Source file: e1088/round9/01_RQ_few_valued_rectangles.md

01. Few-valued real matrices of low rank have large constant submatrices [PROVED]

(Round 9, session jz, 2026-10-18.) This is the "rank-to-rectangle lemma for k-valued low-rank matrices" asked for in target (B). It needs no integrality and loses only 2^{-O_q(sqrt r log r)}.

Statement

Theorem RQ. Let M in R^{m x n} have rank <= r (r >= 1) and let its entries take at most q >= 1 distinct values. Then there are nonempty X' subset [m], Y' subset [n] with |X'| >= 2^{-L} m, |Y'| >= 2^{-L} n, M constant on X' x Y', where L = C q^3 sqrt(r+1) log2(r+q+2) for an absolute constant C.

Literature check (CITED, read 2026-10-18): Hunter-Milojevic-Sudakov-Tomon, arXiv:2411.13510, Thm 1.13: rank-r matrix with NONNEGATIVE INTEGER entries of average <= t has a constant submatrix of size 2^{-O(t sqrt r)}(m,n). That needs integers and degrades with the spread t (values {0,1,N} cost 2^{-O(N sqrt r)}). For q-valued real matrices we found no statement in the literature; RQ follows by combining Rothvoss's hyperplane rounding with a Hadamard-power cleaning step. It may be folklore; we claim no priority.

Ingredients

(J) John's theorem: for a full-dimensional symmetric convex body K in R^s there is an ellipsoid E centred at 0 with E subset K subset sqrt(s) E. [CITED, classical] (S) Sheppard: for unit u, v with = a and g ~ N(0, I_s): P[>= 0 and >= 0] = (1 - arccos(a)/pi)/2 =: h(a). h(0) = 1/4, h'(a) = 1/(2 pi sqrt(1-a^2)) >= 1/(2pi). Hence h(a) >= 1/4 for a >= 0 and h(a) <= 1/4 - b/(2pi) for a <= -b. [CITED, classical] (GL) Cleaning lemma (Gavinsky-Lovett; HMST Lemma 4.7(2)): if B in R^{m x n} has rank <= R and at most mn/(16R) nonzero entries, then B vanishes on some A'' x B'' with |A''| >= m/2, |B''| >= n/2. [CITED; proof recalled: keep rows of degree <= n/(4R) and columns of degree <= m/(4R) in the bipartite graph G of nonzero entries (>= 3m/4, >= 3n/4 survive); a maximal induced matching F of the surviving graph has |F| <= R (it is a nonsingular diagonal submatrix); delete all neighbours of F (<= m/4 rows, <= n/4 columns); what is left spans no edge by maximality. Valid over any field.] (H) Hadamard powers: rank(M^{o j}) <= C(r+j-1, j) (rows are values of degree-j forms in r variables), so for any polynomial P of degree <= q-1, rank P(M) <= sum_{j<q} C(r+j-1, j) = C(r+q-1, q-1) =: R_q (P applied entrywise, M^{o 0} = J).

Lemma R (margin rounding). [PROVED; Rothvoss arXiv:1409.6366, Sec. 3, made quantitative for margins]

Let N in R^{m x n} have rank s and all entries with 1/2 <= |N_xw| <= K. Let Q+ = {N > 0}, Q- = {N < 0} and suppose |Q+| >= delta*mn, 0 < delta <= 1/2. Then there is a rectangle R = X' x Y' with |R| >= (delta/2) 4^{-T} mn and |R n Q-| <= delta |R|, T = ceil(pi K sqrt(s) ln(2/delta^2)). Proof. Write N_xw =

with u_x, v_w in R^s, the u_x spanning. Apply (J) to Ko = conv{+-u_x}; after a linear change u -> A u, v -> A^{-T} v we may assume s^{-1/4} B subset Ko subset s^{1/4} B. Then |u_x| <= s^{1/4}; and for z = s^{-1/4} v_w/|v_w| in Ko, s^{-1/4}|v_w| = <= max_x || <= K, so |v_w| <= K s^{1/4}. Hence the unit vectors satisfy || >= (1/2)/(K sqrt s) =: b with the sign of N_xw. Take independent g_1..g_T ~ N(0,I_s), R_t = {x: >= 0} x {w: >= 0}, R = R_1 n ... n R_T. By (S) and independence: E|R n Q+| >= |Q+| 4^{-T} >= delta mn 4^{-T}; E|R n Q-| <= mn 4^{-T} (1 - 2b/pi)^T <= mn 4^{-T} exp(-2bT/pi) <= (delta^2/2) mn 4^{-T} <= (delta/2) E|R n Q+|. So E[|R n Q+| - |R n Q-|/delta] >= E|R n Q+|/2 >= (delta/2) mn 4^{-T}; fix an outcome achieving this. Then |R n Q-| <= delta |R n Q+| <= delta |R| and |R| >= |R n Q+| >= (delta/2) mn 4^{-T}. [] (Since |X'| <= m and |Y'| <= n, both |X'| >= (delta/2)4^{-T} m and |Y'| >= (delta/2)4^{-T} n.)

Proof of Theorem RQ

Induction on the number q of values present. q = 1: take everything. Let q >= 2, values v_1 < ... < v_q present. Let g = *v_{j+1} - v_j be a largest gap (g* >= (v_q - v_1)/(q-1)), Low = {v_1..v_j}, High = {v_{j+1}..v_q}, theta = (v_j + v_{j+1})/2 and N = (M - theta J)/g*. Then rank N <= r+1, every entry has 1/2 <= |N| <= q-1, and N > 0 exactly on High entries. Replacing N by -N if necessary, |Q+| >= mn/2. Let P be the Lagrange polynomial (degree <= q-1) with P = 1 on the values of the minority side Q- and P = 0 on the majority side; B = P(M) is the 0/1 indicator of Q-, rank B <= R_q by (H). Apply Lemma R with delta = 1/(16 R_q): rectangle X' x Y', |X'| >= (delta/2)4^{-T} m, |Y'| >= (delta/2)4^{-T} n, in which B has at most delta |X'||Y'| = |X'||Y'|/(16 R_q) nonzero entries. By (GL) there is X'' x Y'' (halves) with B = 0, i.e. all entries of M on X'' x Y'' lie on the majority side: at most q-1 values. Loss of this step: log2(m/|X''|) <= 1 + log2(2/delta) + 2T = O(q sqrt(r+1) * log(16 R_q)) = O(q^2 sqrt(r+1) log(r+q+2)), since K = q-1 and log R_q <= (q-1) log2(r+q). The submatrix still has rank <= r. After at most q-1 steps M is constant; total loss O(q^3 sqrt(r+1) log(r+q+2)). []

Remarks

- Only the GAP STRUCTURE at the largest gap enters the rounding (aspect ratio <= q-1), so arbitrary real values (irrational, rational with huge spread, clusters at different scales) are all handled. The cost of cleaning a rank-R_q ~ r^{q-1} indicator is only logarithmic in R_q because the rounding cost is logarithmic in 1/delta.
- For 2^{o(d)}-type applications only L = o(r) matters: RQ stays useful for q up to r^{1/6 - eps}.
- Corollary (additive form). If the entries of M take <= q values on X x Y then also the matrix M + f 1^T + 1 g^T (any f, g) has an ADDITIVE rectangle of the same size (trivially, the same rectangle).
- Earlier partial routes recorded in the scratch analysis (not needed any more): Galois/specialisation (for Q-linear phi on span_Q(values), rank phi(M) <= dim_Q span * r) handles Q-independent values via Sudakov-Tomon; the rational large-spread case was the obstacle; Theorem RQ bypasses it.

Geometric corollary used below

Corollary RQ-G. Let P, Q subset R^D be finite, disjoint, with |{|pq| : p in P, q in Q}| <= q. Then there are R subset P, C subset Q with |R| >= 2^{-L}|P|, |C| >= 2^{-L}|Q| (L as above with r = D+2) and |pq| constant on R x C; consequently

= 0 for p,p' in R, c,c' in C, so dim aff R + dim aff C <= D (dir R, dir C orthogonal). Proof. [|pq|^2] = |p|^2 1^T + 1 |q|^2^T - 2 [] has rank <= D+2; apply RQ; expand |pc|^2 - |p'c|^2 - |pc'|^2 + |p'c'|^2 = -2 = 0. []

Theorem RQ-list (column-listable version). [PROVED - self-checked; see caution]

Let M in R^{m x n} have rank <= r and suppose EACH COLUMN takes at most k distinct values (the value sets may differ from column to column: "k-listable" in the terminology of Singer-Sudan, arXiv:2101.09592). Then there are X' x Y' with |X'| >= 2^{-L} m, |Y'| >= 2^{-L} n, L = C k^3 sqrt(r+1) log2(r+k+2), on which EVERY COLUMN IS CONSTANT. Proof. Induction on k. If at least half of the columns are constant, keep them (all rows): done. Otherwise discard the constant columns (factor 2). For each remaining column j take a largest gap g_j of its value set, threshold theta_j in its middle, and N = (M - 1 theta^T) diag(1/g_j): rank N <= r+1, 1/2 <= |N_ij| <= k-1, and the sign of N_ij says on which side of column j's largest gap the entry lies. Negate N if needed so that Q+ is the majority. Lemma R with delta = 1/(16 R_k), R_k = C(r+k-1, k-1): a rectangle with <= delta minority entries. The minority indicator is B_ij = P_j(M_ij) with a column-dependent Lagrange polynomial P_j of degree <= k-1, so B = sum_{t<k} M^{o t} diag(c_{t,j}) has rank <= R_k; (GL) gives a half-sized sub-rectangle with B = 0. There every column uses only the values on one side of its largest gap: at most k-1 values. Loss per step O(k^2 sqrt(r+1) log(r+k+2)); at most k-1 steps. [] * Theorem RQ (global value set V, |V| = q) follows: a column-constant rectangle whose column values lie in V contains, after keeping the most frequent value, a constant rectangle (extra factor q). * CAUTION. Singer-Sudan (2021; ACM ToCT version) state that their black-box reduction gives only 2^{-O~(d)} for 3-listable matrices and that bounds for "parallel 3-partitioned configurations" would follow from improving Lovett's bound. Theorem RQ-list gives 2^{-O_k(sqrt d log d)} for every fixed k by re-running the margin (hyperplane-rounding) argument with per-column normalisation and paying for the rank blow-up of the cleaning indicator only logarithmically. We checked every step (Lemma R, the rank bound for column-dependent polynomials, the cleaning lemma) but have not seen this statement in print; an independent check is advisable before use outside this project.

Corollary RQ-list-G (equidistant rectangles). Let P, Q subset R^D be finite and disjoint, and suppose every w in Q sees P at most k distinct distances. Then there are R subset P, C subset Q with |R| >= 2^{-L}|P|, |C| >= 2^{-L}|Q| (r = D+2) such that every w in C is equidistant from R; hence dir(aff R) is orthogonal to dir(aff C) and dim aff R + dim aff C <= D. Proof. Columns of [|pw|^2] (indexed by w) are k-listable, rank <= D+2. Column-constant on R x C means |pw|^2 = G(w); expand |pw|^2 - |p'w|^2 - |pw'|^2 + |p'w'|^2 = -2 = 0. [] So for any hereditary class of point sets with size function s'(j) (largest member of affine dimension <= j): if every point of Q sees P in <= k distances then min(|P|,|Q|) <= 2^{L} (s'(floor(D/2)) + 1).

Appendix C. Theorem R (reduction to E1) — round 12

Source file: e1088/round12/01_reduction_to_E1.md

01. Theorem R: the general-n endgame reduces to a class-count statement (E1)

STATUS: Theorem R PROVED (self-checked; referee REFEREE_R.md: CORRECT; uses G1 = LR-4, Lemma H, Caro-Wei only). It BYPASSES E2 (and E3) of round11/02 for every n: the endgame needs ONLY a bound on the number of points with many big classes (E1). Heaviness (G2) is NOT used. E1 itself: see 02_E1.md.

Setting

n >= 3, X rainbow-n-free, affine dimension <= D, N = |X|, s' := s'_n(floor(D/2)). For b in X a CLASS of b is X n S(b,t), t > 0. Fix integers M >= 1 (threshold) and K >= 0. Lambda(b) := {t > 0 : |X n S(b,t)| >= M} (radii of BIG classes of b). Typ := {b in X : |Lambda(b)| <= K} (K-typical points; no other condition). mu_n := 2^{L_k(D+2)} (s' + 1), k = C(n-2,2) + 2(n-2) (G1, round11/02, LR-4 of round11/01). For a finite S in X with S rainbow (all C(|S|,2) distances distinct; automatic if |S| <= 2): W_S := { x in Typ \ S : S u {x} rainbow, and |xs| notin Lambda(s) for every s in S }. (W_emptyset = Typ.)

Tools

(G1) [PROVED, round11/02 + LR-4] For a rainbow (n-2)-set T and Y in the T-generic points: the number of ordered label-disjoint pairs in Y is <= 4^{n-2} mu_n |Y|. Degree form: if every value v has deg_Y(v) := #{x in Y : v in L_x} <= |Y|/(4(n-2)), then |Y| <= 2 * 4^{n-2} mu_n + 4. [Pairs sharing a value <= sum_v deg^2 <= maxdeg * (n-2)|Y| <= |Y|^2/4; so >= |Y|(|Y|-1) - |Y|^2/4 >= |Y|^2/2 - ...] (H) [PROVED, round 8 Lemma H, symmetric form] F_x (x in I) in a space of functions of dimension m, F_x(x) != 0, and for all x != y in I: F_x(y) = 0 or F_y(x) = 0. Then |I| <= m^2. (CW) [CITED] Caro-Wei: average degree dbar => independent set of size >= |V|/(dbar+1). (DIM) polynomials of degree <= e in the D+1 lifted coordinates (|y|^2, y_1..y_D): dimension C(D+1+e, e).

Theorem R

Put e_r := K + 2r + C(r,2), P_r := C(D+1+e_r, e_r)^2 and define m_{n-2} := max(4(n-2)^2 M , 2 * 4^{n-2} mu_n + 4), m_r := P_r * (2 m_{r+1} + 2 r^2 M + 1) (r = n-3, ..., 0). Then |W_S| <= m_{|S|} for every rainbow S in Typ with |S| <= n-2. In particular |Typ| <= m_0 <= (D+2)^{O(n K + n^3)} (M + 4^n mu_n). (Points of S need not be typical for the base case; for r < n-2 only the NEW points b must be in Typ.)

Proof

Base r = n-2. Y := W_S consists of S-generic points. If x in Y and |xs| = v then v notin Lambda(s), so |X n S(s,v)| < M. Hence deg_Y(v) <= sum_{s in S} |Y n S(s,v)| < (n-2) M. If |Y| >= 4(n-2)^2 M the degree form of (G1) applies: |Y| <= 2*4^{n-2} mu_n + 4. So |Y| <= m_{n-2}.

Step r < n-2. W := W_S, |S| = r. For b in W (so b in Typ, S u {b} rainbow, |bs| notin Lambda(s)) put F_b(y) := prod_{lam in Lambda(b)} (|y-b|^2 - lam^2) * prod_{s in S} (|y-b|^2 - |y-s|^2) * prod_{s in S} (|y-b|^2 - |b-s|^2) * prod_{d in D_S} (|y-b|^2 - d^2). Each factor has degree 1 in lifted coordinates (the bisector factor is -2 + |b|^2 - |s|^2), at most K + 2r + C(r,2) = e_r factors: F_b lies in a space of dimension C(D+1+e_r, e_r). F_b(b) != 0 (factors -lam^2, -|bs|^2, -|bs|^2, -d^2, all nonzero). Claim: # {x in W \ {b} : F_b(x) != 0} <= m_{r+1} + r^2 M. Let x in W \ {b}, F_b(x) != 0, so |xb| notin Lambda(b) u {|xs|} u {|bs|} u D_S. If moreover |xs| != |bs'| for all s, s' in S, then all distances of S u {b,x} are distinct (D_S u {|bs|} and D_S u {|xs|} are rainbow, the two label sets are disjoint, |xb| is new), so x in W_{S u {b}} (x in Typ, |xs| notin Lambda(s), |xb| notin Lambda(b)); there are <= m_{r+1} such x. Otherwise x in S(s, |bs'|) for some s, s'; since |xs| notin Lambda(s), this class of s is not big: < M points; r^2 pairs (s,s'). Unkilled graph G on W: {x,b} an edge iff F_b(x) != 0 and F_x(b) != 0. By the claim e(G) <= |W|(m_{r+1} + r^2 M), so dbar <= 2(m_{r+1} + r^2 M). (CW) gives an independent set I, |I| >= |W|/(dbar+1); every pair of I is killed from one side; (H): |I| <= P_r. So |W| <= P_r (2 m_{r+1} + 2 r^2 M + 1) = m_r. []

Corollary R1 (what remains)

N <= m_0 + |X \ Typ|. So f_D(n) = 2^{o(D)} follows (with the halving recursion of n4_full_proof Section 9) from: (E1) for some M <= 2^{o(D)} (s'+1) and K = o(D/log D): at most N/2 points of X have more than K classes of size >= M (or any other bound |X \ Typ| <= N/2 that holds whenever N > 2 m_0). Cost check: P_r = 2^{O((K + r^2) log D)}; the product over n-1 levels is 2^{O(n(K + n^2) log D)} = 2^{o(D)} when K = o(D/log D). LR enters ONLY through mu_n (the base). No GL, no heaviness, no Prop 4.6-4.8 analogues needed.

Corollary R2 (n = 4 rederived, new short endgame) [PROVED given round11 Step 3]

For n = 4, Props 4.7 + 4.8 of n4_full_proof give |{points with >= 2 classes of size >= M}| <= mu4(2). With K = 1: N <= mu4(2) + m_0, m_0 = poly(D)(M + mu_4). This replaces Steps 2 and 4-7 (heaviness, bisector charging, equal labels). (Prop 4.8 needs N >= 32 C(D+3,2)(mu3+3); if that fails N is already small.)

Sanity checks

- n = 3, M = 1: W_(b) is empty, so the argument gives |Typ| <= C(D+1+K,K)^2 (Lemma H itself). OK.
- s-distance sets: consistent (N = poly(D)); they impose NO lower bound on K in E1 (an earlier claim to the contrary was withdrawn after the referee pass).
- Theorem R does not care whether points of S are typical; it is a statement about the "small-class" relation only.

Appendix D. Theorem R+, HR, dyadic regularity — round 12

Source file: e1088/round12/03_two_class_levels.md

03. Theorem R+: two-big-class columns are harmless at every level r >= n-4

STATUS: PROVED (self-checked; uses G1, Lemma S2 of file 02, GL, LR, (OR), Lemma H, CW). Consequences: * n = 4: a self-contained endgame (no heaviness, no Steps 4-7, no Prop 4.8); f_d(4) = 2^{O(sqrt d log d)} again. * n >= 5: E1 is needed ONLY at the top n-4 levels (r < n-4). For n = 5: only at r = 0, i.e. for points of X itself; and every point of X has at most m_1 points outside its big classes.

Scheme (relative classes, increasing thresholds)

Thresholds tau_0 < tau_1 < ... < tau_{n-3}, tau_r := Q^r tau_0, Q := (C0 n (D+2)^{k'})^{12} (C0 absolute; Q = poly(D) for fixed n; 2^L is NOT needed in Q) with k' := C(n-2,2) + 2n (types in Step A below), tau_0 >= tau_min := 2^{L_{k'}+22} 4^n Q^{1/2} (D+2)^{4k'} (mu_n + s' + 1) [Q^{1/2} factor: referee fix F1]. Convention tau_{-1} := tau_0/Q (F3). Objects are SEQUENCES S = (s_1..s_r) (s_k chosen at level k-1), with sets W_S defined recursively: W_() = X. For b in W_S (level r = |S|): R_b := {|bs| : s in S} u D_S; Big_b := {lam notin R_b : |S(b,lam) n W_S| >= tau_r}. Case B: |Big_b| <= 1. Case A: |Big_b| >= 2 (pick two radii rho_i, rho_j in Big_b; C_i := S(b,rho_i) n W_S, C_j). W_{S,b} := {x in W_S \ {b} : S u {b,x} rainbow, |xb| notin Big_b} (case B only). Invariant: every class of s_k meets W_S (|S| >= k) in < tau_{k-1} points. [x in W_{S,b} has |xb| notin Big_b u R_b.] Kill polynomials: F_b := Delta_b * prod_{lam in Big_b} (|y-b|^2 - lam^2) in case B, F_b := Delta_b (|y-b|^2-rho_i^2) (|y-b|^2-rho_j^2) in case A, Delta_b := prod_s (|y-b|^2-|y-s|^2) prod_s (|y-b|^2-|bs|^2) prod_{d in D_S} (|y-b|^2-d^2). deg F_b <= 2 + 2r + C(r,2); F_b(b) != 0.

Theorem R+

Let r0 := max(0, n-4). Suppose that at every level r < r0 every column has <= K radii in Big_b (case B_K; E1, see below); then use the K-label kill there (P_r with K in place of 1). Then |W_S| <= m_r for all S, where m_{n-2} := max(4(n-2)^2 tau_{n-3}, 2*4^{n-2} mu_n + 4) and m_r := P_r (2 max(m_{r+1}, e_A) + 2 (r^2 + 2n) tau_{r-1} + 1), e_A := 2^{L_{k'}+4}(s'+1), P_r := C(D+3+2r+C(r,2), 2+2r+C(r,2))^2. In particular N = |W_()| <= m_0 <= D^{O_n(1)} tau_0 = 2^{O_n(sqrt D log D)} (s'+1) (tau_0 := tau_min).

Proof

Base and case B: exactly as Theorem R (file 01), with "class < M" replaced by "class meets W_S in < tau_{k-1}" (the invariant): base degrees < (n-2) tau_{n-3}; case B: x in W_S{b} with F_b(x) != 0 lies in W_{S,b} or on one of <= r^2 spheres S(s,|bs'|) meeting W_S in < tau_{r-1} points. Step A (r >= n-4, column b in case A). [Referee fixes F1, F2 applied: A_0 is taken inside W_S \ {b}; the small-P branch uses theta_r := Q^{-1/2} tau_r, and for |P|, |Q| >= theta_r the S2 count is done directly: the S'-coincidences cost <= n(n-4) tau_{r-1}(1/|P| + 1/|Q|) <= 2 n^2 Q^{-1/2} and G1 costs <= 2*4^{n-2} mu_n

Q^{1/2}/tau_r, so eta <= 2^{-20}(D+2)^{-4k'} and eta^{1/4} <= 1/(32 (D+2)^{k'}).] Fix S' = any n-4 entries of S (spread on W_S: classes < tau_{r-1} <= tau_r / Q <= |C_i|/Q). Let A_0 := {a in W_S : a notin C_i u C_j, Delta_b(a) != 0, a on none of the r^2 spheres S(s,|bs'|), a notin S(s,rho_i) u S(s,rho_j) (s in S')}. For a in A_0, T := {b, a} u S' is rainbow and rho_i, rho_j notin D_T. Non-generic points of C_i for T lie on <= C(n-2,2)+1 spheres about a (radii in D_T u {rho_i}), on the bisectors H(a,s) (s in S'), or on small spheres of s in S' (< n^2 tau_{r-1} points); the bisectors H(s,s') and spheres S(s,d), d in D_S, contain no point of W_S (S u {x} rainbow). Lemma S2 (file 02) with c = 2/Q and the four-case analysis (below) give: a sees C_i or C_j, up to a fraction eta' <= 1/(16 (D+2)^{k'}), on <= k' types: spheres about a (column functions/constants) and |xa|^2 = |xs|^2 (row functions). Split A_0 = A_i u A_j accordingly, say |A_i| >= |A_0|/2. The exceptions in C_i x A_i are the support of the product of the k' type differences (rank <= (D+2)^{k'}, density <= eta'): (GL) gives C'' x A'' (halves), then LR (k' types, rank D+2) a single-type rectangle R x C', |R| >= 2^{-L} |C_i|/2 > s', and every type gives (OR); so |C'| <= s' and |A_0| <= 2(2^{L+1} s' + 1) <= e_A. So #{x in W_S \ {b} : F_b(x) != 0} <= e_A + r^2 tau_{r-1} + 2 n tau_{r-1}. Four-case analysis (from S2): with p, q the distributions of |xa| on the generic parts P, Q and 1 - eta <= P(|xa| = |wa| or |xa| = rho_j or |wa| = rho_i): if (1-p_{rho_j})(1-q_{rho_i}) >= sqrt(eta), the conditional distributions agree on one value t up to 2 sqrt(eta): P on {rho_j, t}, Q on {rho_i, t}; otherwise P on S(a,rho_j) or Q on S(a,rho_i) up to eta^{1/4}. If |P| or |Q| < theta := 2*4^{n-2} mu_n eta^{-1}, that class is covered by the non-generic loci up to theta + O(n^2 tau_{r-1}) points, a fraction <= 1/Q^{1/2}. Then the Lemma H + Caro-Wei step at level r is as in Theorem R with max(m_{r+1}, e_A) in place of m_{r+1}. []

Corollary n = 4 [PROVED]

r0 = 0: no assumption. N <= m_0 = 2^{O(sqrt D log D)}(s'+1), hence f_d(4) = 2^{O(sqrt d log d)} (Section 9 of n4_full_proof). Ingredients: G1 (LR, k = 5), S2 (= Prop 4.6), GL, LR (k' types), OR, Lemma H, Caro-Wei.

Corollary n = 5 [PROVED part]

Levels 1, 2, 3 need nothing. For EVERY b in X put W_b := points in classes of b meeting X in < tau_0 points; then |W_b| <= m_1 (level-1 columns are spread-tested against b itself). So: *() every point of X has all but m_1 points of X in its classes of size >= tau_0. Level 0 with K labels: columns with <= K classes of size >= tau_0 may use F_b of degree K (P_0 = C(D+1+K,K)^2), so N <= C(D+1+K,K)^2 (2 m_1 + 1) + #{b : b has > K classes of size >= tau_0}. Dyadic regularity [PROVED from ()* applied at every threshold tau >= tau_min]: m_1(tau) <= Gamma tau with Gamma = m_1/tau_0 <= P_1 P_2 * 200 Q^2 = D^{O(1)} for fixed n (n = 5: Gamma ~ D^{334}; referee's count), so for EVERY point b and every tau >= tau_min, the classes of b of size < tau contain <= Gamma tau points; hence b has <= 2 Gamma classes with size in [tau, 2 tau), and the classes of size

> = N/(2 Gamma) contain at least half of X (unconditional heaviness for n = 5, without G2). Lemma HH (hereditary heaviness, n = 5) [PROVED]: for every a in X and every V in X \ {a} with |V| >= 2 Gamma tau_min, a has a class containing >= |V|/(2 Gamma) points of V. [Run levels 1-3 with S = (a), ground set V, tau_0' := |V|/(2 Gamma): if every class of a met V in < tau_0' then |V| <= Gamma tau_0' = |V|/2.] General n (same proof, start at level n-4): for every rainbow (n-4)-set S and every set W of S-generic points with |W| >= 2 Gamma_n tau_min, some s in S has a class containing >= |W|/(2 Gamma_n) points of W, Gamma_n = D^{O_n(1)}. (n = 4: S empty, i.e. N <= 2 Gamma_4 tau_min, the theorem.) Why HH does not close n = 5: Step A at level 0 needs a point that is (1/Q)-SPREAD on two big classes of b; HH gives every point a (1/(2 Gamma))-heavy class, and Gamma >= Q^2 > Q is forced by the

threshold escalation, so HH neither produces nor excludes spread points. Missing (E1 at r = 0): a bound on the points with two classes of size >= tau_0 that can be used in the level-0 Lemma H, e.g. "at most N/2 points have more than K = o(D/log D) classes of size >= tau_0". Points with <= 1 such class are one-sphere points by (*), at most (D+2)^2 (2 m_1 + 3) of them (Lemma H with F_b = |y-b|^2 - lam_b^2): so in a counterexample almost every point has >= 2 big classes.

General n (restated after referee item 5)

R+ as written assumes RELATIVE case B_K at levels r < n-4. To need only the ABSOLUTE E1 of file 02 (for points of X), use Theorem R's sets W_S (inside Typ, threshold M) at levels r < n-4 and switch to the R+ scheme at level n-4 with tau_{n-5} := M, tau_r := Q^{r-n+5} M; the invariant "classes of s meet W_S in < M" then holds. Checked by the referee, written here only in sketch form [SKETCH: referee-checked, not written in full; treat as unproved for n >= 6 until written]. Granting it: E1 is needed exactly at levels r = 0..n-5, where S has fewer than n-4 points and Lemma S2 has no spread points available. At those levels a case-A column must be shown rare or be killed by a bounded-degree polynomial. This is the precise remaining gap.

Appendix E. Referee fixes F1–F3 to Theorem R+ — round 12

Source file: e1088/round12/REFEREE_R.md

Referee report on round 12 (Theorem R, Lemma S2, Theorem R+, n = 4 and n = 5 corollaries)

Every step below was re-derived by hand. Sanity scripts: a local sanity-check script (not included).

item	verdict
1. Theorem R	CORRECT
2. Lemma S2 / Step A	S2 CORRECT. Step A: CORRECT WITH FIXES (one real but repairable threshold gap, F1)
3. n = 4 corollary	CORRECT WITH FIXES (F1–F3). The recursion closes
4. n = 5: (*), dyadic regularity, HH	CORRECT (after F1). No typicality is used. Gamma's exponent is misstated
5. Gap / labelling	Gap correct for n = 5. Hypothesis for n >= 6 needs restating. Three items in 02 are labelled PROVED but should not be

1. Theorem R: correct

- **Factors and F_b(b).** All factors are degree 1 in (|y|^2, y). The bisector factor is -2 + |b|^2 - |s|^2. There are at most e_r factors. At y = b the factors are -lam^2, -|bs|^2, -|bs|^2 and -d^2, all nonzero.
- **Claim.**
- If {|xs|} n {|bs'|} = {}, then S u {b,x} is rainbow: D_S u {|bs|} and D_S u {|xs|} are rainbow, the two label sets are disjoint, and |xb| is new. So x is in W_{S u b} (x in Typ, |xb| notin Lambda(b)).
- Otherwise |xs| = |bs'| is notin Lambda(s), so x lies in a small class: at most r^2 (M-1) such x.
- **Base.** A value v in Lambda(s) is never an s-label of a point of W_S. So deg(v) < (n-2)M <= |Y|/(4(n-2)) once |Y| >= 4(n-2)^2 M. The degree form checks: |Y| <= (4/3)(4^{n-2} mu_n + 1).
- **Caro-Wei + Lemma H.** Correct. e(G) is overcounted by a factor 2, which is harmless.
- **Circularity.** None. Points of S need not be typical.
- **Cosmetic.**
- "Settles E2/E3" should read "bypasses".
- The n = 3 sanity line holds through the argument (W_(b) is empty), not through the formula for m_0.

2. Lemma S2 and Step A

S2: correct. - Coincidences involving some a_p with p >= 2 come from (n-2)^2 - 4 = n(n-4) ordered label pairs. Each costs <= c, and n(n-4) <= 2(n-3)^2. - The three surviving events are exactly the ones stated. - G1 bounds the label-disjoint cross pairs.

Four-case analysis: correct. - After conditioning: sum p'q' >= 1 - sqrt(eta) gives p'_{t} >= *1 - sqrt(eta) and q'_{t}* >= 1 - 2 sqrt(eta). - Otherwise one mass is < eta^{1/4}.

T rainbow, rho_i, rho_j notin D_T: correct. These follow from the A_0 conditions.

Genericity bookkeeping: correct. - Every x in W_S has S u {x} rainbow, by induction through W_{S,b}. So H(s,s') and S(s,d) with d in D_S miss W_S. - A non-generic x in C_i lies in one of these places: - S(a,d) with d in D_T u {rho_i}; - H(a,s) for s in S'; - one of (n-4)(2n-6) < n^2 classes of s in S' (radii rho_i, |bs'|, |as'|, |ab|). - Computer check: 70,213 random grid configurations (n = 4, 5, 6), 20,361 non-generic, 0 outside the listed loci.

Exceptions = support of the Hadamard product: correct. - The type list is fixed: C(n-2,2) + n - 1 <= k' types, with t*(a) as a column function. - Each factor has rank <= D+2, so the product has rank <= (D+2)^{k'}. - Rows and columns are disjoint. - Every type gives OR. Since |R| > s', we get |A_0| <= 2^{L+2} s' + 2 <= e_A.

F1 (real gap, repairable). - S2 is used with c = 2/Q, which needs |P| >= tau_r/2 (spread relative to the generic part). - The fallback only covers |P| < theta = 2*4^{n-2} mu_n / eta, and theta << tau_r. - For theta <= |P| < tau_r/2 neither branch applies. Up to half of C_i is then uncontrolled, and the GL density fails. - (Also, theta/tau_0 <= Q^{-1/2} silently needs 2^{L_{k'}} >= poly(D) Q^{1/2}.)

Fix: - Set theta_r := Q^{-1/2} tau_r. - If |P| < theta_r: C_i is covered by the non-generic loci up to a fraction < Q^{-1/2} + n^2/Q. - Otherwise count directly. The S'-coincidences cost <= n(n-4) tau_{r-1} (1/|P| + 1/|Q|) <= 2n^2 Q^{-1/2}. G1 costs <= 2*4^{n-2} mu_n Q^{1/2} / tau_r. - We need eta <= 2^{-20} (D+2)^{-4k'}. This holds for C0 >= 16 and tau_0 >= 2^{22} 4^n Q^{1/2} (D+2)^{4k'} mu_n, i.e. multiply tau_min by Q^{1/2} = poly(D). - Asymptotics are unchanged, and Gamma (a ratio) is unaffected.

F2. At r = 0, A_0 contains b itself. Take A_0 inside W_S \ {b}.

F3. tau_{-1} is undefined in m_0. Its coefficient should be 0 when S' is empty.

n = 5 dichotomy (file 02), part (i): overstated (not used downstream). - With c = 1/(64(D+2)^7), the eta^{1/4} loss is far above the claimed 1/(16(D+2)^7). One needs c ~ (D+2)^{-4k'}. - "Sees V" should read "sees C_1 or C_2". - The conclusion about X ignores a_1 in C_1 u C_2, where rho_i is in D_T and S2 does not apply.

3. n = 4: correct with fixes (F1–F3)

- r0 = 0, so nothing is assumed.
- Ingredients: G1 (k = 5), S2 (with S' empty, so only the G1 error term), GL, LR (k' = 9), OR, Lemma H (degrees 2 and 4) and Caro-Wei. No heaviness or Props 4.7/4.8 are needed.
- I checked level 1 (S' empty, S(s_1, |bs_1|) excluded from A_0) and the base (G1 with T = {s_1, s_2}).
- Thresholds after F1:
- tau_0 = 2^{L_9(D+2)} poly(D) (mu_4 + s' + 1), with mu_4 = 2^{L_5(D+2)} (s'+1);
- Q = poly(D);
- m_0 = poly(D) Q^{O(1)} tau_0.
- So s'(D) <= 2^{C sqrt(D) log D} (s'(floor(D/2)) + 1). Summing over the halvings gives f_d(4) = 2^{O(sqrt(d) log d)}. The recursion closes.

4. n = 5: correct

- **(*).** Put W_(b) := the points in classes of b of size < tau_0.
- R_b is empty.
- Levels 1–3 use only the invariant "classes of b meet W_(b) in < tau_0".
- Case A at level 1 uses S' = {b}, whose spread is exactly that invariant.
- All spheres excluded from A_0 are classes of b.

No typicality is used, so *() holds for every point. (Example check: X = Y u {o} with o generic gives |Y| <= m_1. That is consistent, not a contradiction.) - Level 0 with K labels: correct. - Dyadic regularity: correct. m_1(tau)/tau is nonincreasing in tau, because each term has the form max(a tau, const). - Gamma's exponent is wrong. With k' = 13, Q ~ D^{156}, so Gamma ~ P_1 P_2 Q^2 ~ D^{334}, not D^{210} (cosmetic). - Lemma HH: correct. The levels never use points outside the ground set, and tau' >= tau_min is assumed. - General-n HH: correct.* At levels >= n-4 only the tau_{r-1} bound on the classes of S is used.

5. Remaining gap and labelling

- **n = 5.** Correctly identified: what is missing is a bound on the points of X with more than K = o(D/log D) classes of size >= tau_0. Dyadic regularity only gives K <= 2 Gamma log(N/tau_0) = poly(D) log N, which is too large.
- **n >= 6.** As written, R+ assumes that EVERY column at levels r < n-4 has <= K RELATIVE big radii (relative to W_S, threshold tau_r). That is not the E1 of file 02, which is absolute and about most points.
- Fix (checked, not written): use Theorem R's absolute sets W_S inside Typ at levels < n-4, then switch to R+ at level n-4 with tau_{n-5} := M. The invariant "classes < M" holds there.
- Until this is written up, "E1 only at the top levels" should not be asserted for n >= 6.
- **Mislabelled PROVED in 02:**
- **(C1) is wrong.**
    - s-distance sets have N <= C(D+s,s) = poly(D) < M <= m_0, so they never meet E1's hypothesis N > 2m_0.
    - Their classes are unbalanced. For example, Johnson with D+1 = 100, s = 5 has classes of 77%, 21%, 1.8%, 0.06% of N.
    - R2 itself uses K = 1 for n = 4.
    - "E1 must allow K >= C(n,2)-1" should be deleted in 01 and 02. The only supported lower bound is K >= 2 for n >= 6.
- **(C3)–(C5) are heuristic.**
- **The n = 5 dichotomy** is overstated (section 2).
- **Everything else** marked PROVED in 01 and 03 holds, subject to F1–F3. Nothing claims f_d(n) = 2^{o(d)} for n >= 5.

Appendix F. E1 for n = 5 — round 13

Source file: e1088/round13/01_E1_n5.md

01. E1 for n = 5: every point has at most 11 classes of size >= M [CANDIDATE PROOF, self-checked in 02]

STATUS: PROVED (candidate; written in full below; adversarial self-check in 02_selfcheck.md). Uses only: G1 (round12/01 Tools = round11/02 + LR-4), HR (hereditary form of round12/03 (*) + dyadic regularity, see below), GL, RL (= LR with column types only), OR, HP. No heaviness, no spread points, no Lemma S2.

Setting and notation

X rainbow-5-free, affine dimension <= D, s' := s'_5(floor(D/2)), mu := mu_5 = 2^{L_9(D+2)} (s'+1). L_k := L_k(D+2) from Theorem LR (nondecreasing in k). For c in X, a CLASS of c is X n S(c,t), t > 0. S(c,U) := union of S(c,u), u in U. H(p,q) := perpendicular bisector. For a rainbow triple T = {t1,t2,t3}: D_T its 3 distances, Y_T = T-generic points (x notin T, T u {x} rainbow), L_x = {|x t1|, |x t2|, |x t3|}. (G1) [CITED, round12/01 Tools; round11/02 + LR-4] For P, Q in Y_T the number of (x,w) in P x Q with L_x n L_w = {} is <= 64 mu (|P| + |Q|). (64 = 4^{n-2}; ordered label-disjoint pairs in Y = P u Q.) (HR) [CITED, round12/03: (*) run with ground set V, dyadic regularity, Lemma HH proof; referee12 item 4] There are Gamma = D^{O(1)} and tau_min = 2^{O(sqrt D log D)}(s'+1) such that for every a in X, every V in X \ {a} and every theta >= tau_min: #{x in V : |V n S(a,|ax|)| < theta} <= Gamma theta. (GL), (RL), (OR), (HP): as in round11/n4_full_proof Section 1. e_w := 2^{L_w+1} (s'+1).

Parameters: theta_abs := max(tau_min, 2^{L_8+1}(s'+1) + 1); theta_rel(w) := 1/(16 Gamma (D+2)^w); eps := 1/(768 Gamma (D+2)^9); M := max(3 * 2^{23} Gamma^2 (D+2)^{24} mu, 16 (D+2)^8 Gamma theta_abs, 960 (D+2)^2 e_8, 2^{L_8+2}(s'+1)). All are 2^{O(sqrt D log D)} (s'+1).

Theorem E1(5)

Every b in X has at most 11 classes of size >= M. Consequently (Theorem R of round12/01 with K = 11, Typ = X): |X| <= m_0 = (D+2)^{O(1)} (M + 4^5 mu), i.e. s'_5(D) <= 2^{C sqrt(D) log(D+2)} (s'_5(floor(D/2)) + 1), and (halving recursion, n4_full_proof Section 9) f_d(5) = 2^{O(sqrt d log d)}.

Proof

Suppose b has classes C_i = X n S(b, rho_i), i in I, |I| = K >= 12, |C_i| >= M, rho_i distinct. Fix l != m in I. Throughout a1 in C_l, a2 in C_m, T := {b, a1, a2}, D_T = {rho_l, rho_m, |a1 a2|}.

Step 1 (trichotomy without spread) [PROVED]

Let k != k' in I \ {l,m} and |a1 a2| notin {rho_l, rho_m, rho_k, rho_k'} ("(k,k')-valid"). Then T is rainbow and rho_k, rho_k' notin D_T. For x in C_k: x is T-generic iff |x a1|, |x a2| notin D_T u {rho_k} and |x a1|

!= |x a2|; so C_k \ Y_T lies in S(a1, D_T u {rho_k}) u S(a2, D_T u {rho_k}) u H(a1,a2). (1) Put P := C_k n Y_T, Q := C_k' n Y_T, Q' := {w in Q : rho_k notin {|w a1|, |w a2|}}. (i) If |Q'| >= eps |C_k'|: by (G1) some w0 in Q' has at most ex := 64 mu (|C_k| + |C_k'|)/(eps |C_k'|) label-disjoint partners x in P. For any other x in P, L_x = {rho_k, |xa1|, |xa2|} meets L_w0 = {rho_k', |w0 a1|, |w0 a2|}; rho_k != rho_k' and rho_k notin L_w0, so |x a_i| in V := L_w0 for some i. With (1): C_k in S(a1,W) u S(a2,W) u H(a1,a2) u Ex, W := L_w0 u D_T u {rho_k} (|W| <= 7), |Ex| <= ex. Since |C_k|, |C_k'| >= M: |Ex| <= delta1 |C_k|, delta1 := 128 mu/(eps M). DEFINE Good_k(a1,a2): C_k is covered as displayed for some W (|W| <= 7) and Ex (|Ex| <= delta1 |C_k|). (ii) Otherwise |Q'| < eps |C_k'|. Points of C_k' \ Q are covered by (1) (with k'), points of Q \ Q' lie in S(a1,rho_k) u S(a2,rho_k). With Lam_k'(a1,a2) := S(a1, D_T u {rho_k'}) u S(a2, D_T u {rho_k'}) u H(a1,a2): | C_k' \ (Lam_k' u S(a1,rho_k) u S(a2,rho_k)) | < eps |C_k'|. (ii) *{k,k'} So for every (k,k')-valid pair: Good_k(a1,a2) or (ii)*{k,k'}.

Step 2 (summing over k) [PROVED]

Fix a1, a2, k'. Let J be a set of indices k (k notin {k',l,m}, distinct radii) with (ii)*{k,k'} for all k in J. If |J| >= 3 then |C_k' \ Lam_k'(a1,a2)| < 3 eps |C_k'|. Proof. c := |C_k' \ Lam_k'|. Each k in J puts > c - eps|C_k'| points of C_k' \ Lam_k' on S(a1,rho_k) u S(a2,rho_k). Spheres about a1 with distinct radii are disjoint, likewise about a2, so the sum over k in J is <= 2c: |J|(c - eps|C_k'|) < 2c (or c <= eps|C_k'| outright), i.e. c (|J|-2) < |J| eps |C_k'|, c < 3 eps |C_k'|. [] DEFINE Cov_k'(a1,a2): |C_k' \ Lam_k'(a1,a2)| < 3 eps |C_k'|. (Lam_k' = S(a*{1,2}, W') u H(a1,a2), W' = {rho_l, rho_m, |a1a2|, rho_k'}, |W'| = 4.)

Step 3 (two-centre lemma) [PROVED]

Let j in I \ {l,m}, a2 in C_m fixed, w in {4,7}, and Z in C_l a set of a1 such that, for each a1 in Z, C_j in S(a1,W(a1)) u S(a2,W(a1)) u H(a1,a2) u Ex(a1), |W(a1)| <= w, |Ex(a1)| <= delta |C_j|, with delta <= theta_rel(w)/(16 (D+2)^{w+1}). Put theta := max(theta_rel(w) |C_j|, theta_abs). Then either |Z| <= (w+1) e_{w+1}, or a2 is w-FEW on C_j: all but Gamma theta <= |C_j|/(16 (D+2)^w) points of C_j lie on at most w spheres about a2. Proof. For a radius lam put R_lam := C_j n S(a2,lam) and Z_lam := {a1 in Z : lam notin W(a1)}. For x in R_lam and a1 in Z_lam: x notin S(a2,W(a1)) (as |x a2| = lam notin W(a1)); x in H(a1,a2) iff |x a1| = lam. Hence every x in R_lam \ Ex(a1) has |x a1| in W(a1) u {lam}. Claim: if |R_lam| >= theta then |Z_lam| <= e_{w+1}. Rows R_lam, columns Z_lam (disjoint: C_j n C_l = {}). B[x,a1] := prod_{v in W(a1) u {lam}} (|x a1|^2 - v^2) (padded to w+1 factors): each factor |x|^2 + |a1|^2 - v^2 - 2 has rank <= D+2, so rank B <= (D+2)^{w+1} (HP); B != 0 only on Ex(a1), <= delta|C_j| <= (delta/theta_rel(w)) |R_lam| <= |R_lam|/(16 (D+2)^{w+1}) per column. (GL): R'' x Z'' (halves) with B = 0, i.e. every a1 in Z'' sees R'' in <= w+1 distances. (RL, k = w+1): R''' x Z''' with every a1 in Z''' equidistant from R''', |R'''| >= 2^{-L_{w+1}} |R''|, |Z'''| >= 2^{-L_{w+1}} |Z''|. (OR): min(|R'''|,|Z'''|) <= s'. |R'''| >= 2^{-L_{w+1}} (theta_abs - 1)/2 > s', so |Z'''| <= s' and |Z_lam| <= 2^{L_{w+1}+1} s' + 1 <= e_{w+1}. If a2 has w+1 distinct radii lam_1..lam_{w+1} with |R_lam_i| >= theta, every a1 in Z lies in some Z_lam_i (|W(a1)| <= w), so |Z| <= (w+1) e_{w+1}. Otherwise at most w classes of a2 meet C_j in >= theta points, and by (HR) (V = C_j, a2 notin C_j, theta >= tau_min) the remaining points of C_j number <= Gamma theta; and Gamma theta <= max(|C_j|/(16(D+2)^w), Gamma theta_abs) <= |C_j|/(16 (D+2)^w) as |C_j| >= M >= 16(D+2)^8 Gamma theta_abs. []

Step 4 (few w-few centres) [PROVED]

For j in I \ {m}: the set F of a2 in C_m that are w-few on C_j has |F| <= e_w. Proof. Rows C_j, columns F; B[x,a2] := prod_{v in W(a2)} (|x a2|^2 - v^2) (w factors), rank <= (D+2)^w, <= |C_j|/(16(D+2)^w) nonzeros per column. (GL), (RL, k = w), (OR) as in Step 3; |R| >= 2^{-L_w}|C_j|/2 > s' (M >= 2^{L_8+2}(s'+1)), so |F| <= 2^{L_w+1} s' + 1 <= e_w. [] Corollary (Steps 3+4). For j in I \ {l,m} and

Z(a2) as in Step 3 for each a2 in C_m: sum_{a2 in C_m} |Z(a2)| <= e_w |C_l| + (w+1) e_{w+1} |C_m|. Applications: (a) Z(a2) := {a1 : Good_j(a1,a2)}, w = 7, delta = delta1 <= theta_rel(7)/(16(D+2)^8) = 1/(256 Gamma (D+2)^15) [by M >= 3*2^23 Gamma^2 (D+2)^24 mu]: #{(a1,a2) : Good_j} <= e_7|C_l| + 8 e_8 |C_m|. (b) Z(a2) := {a1 : Cov_j(a1,a2)}, w = 4, delta = 3 eps = theta_rel(4)/(16 (D+2)^5):

{(a1,a2) : Cov_j} <= e_4 |C_l| + 5 e_5 |C_m|.

Step 5 (counting; contradiction) [PROVED]

V_lm := {(a1,a2) in C_l x C_m : |a1 a2| notin {rho_l, rho_m}}. For (a1,a2) in V_lm at most one k *in I has rho_k* = |a1a2|; put I0 := I \ {l, m, k*}, |I0| >= K-3; every (k,k') with k != k' in I0 is valid. g := #{k in I0 : Good_k(a1,a2)}. If g <= |I0| - 4, then for every k' in I0 the set J := {k in I0 \ {k'} : not Good_k} has |J| >= 3 and (ii)_{k,k'} for k in J (Step 1), so Cov_k' holds (Step 2) for all k' in I0. Hence g + #{k' : Cov_k'} >= K - 6 on V_lm, and by the Corollary (K-6) |V_lm| <= (K-2) [(e_7 + e_4)|C_l| + (8 e_8 + 5 e_5)|C_m|] <= 2 * 13 e_8 (|C_l| + |C_m|) (K >= 12). So B[a1,a2] := (|a1a2|^2 - rho_l^2)(|a1a2|^2 - rho_m^2) on C_l x C_m (rank <= (D+2)^2, support V_lm) has density <= 26 e_8 (1/|C_l| + 1/|C_m|) <= 52 e_8/M <= 1/(16(D+2)^2). (GL): C_l" x C_m" (halves) with all distances in {rho_l, rho_m}; (RL, k = 2 constant types): R x C with every a2 in C equidistant from R, |R| >= 2^{-L_2}|C_l|/2 > s', |C| >= 2^{-L_2}|C_m|/2 > s'; contradicts (OR). []

Remarks

- Nothing about b beyond "K >= 12 classes of size >= M" is used; the bound holds for EVERY point (no "most points" caveat). Round12's (C2) remark (X = Y u {0}, Y rainbow-4-free, "0 can have |Y|/M classes of size M") was never verified; for M as defined here the theorem says it is false whenever |Y|/M >= 12. No known fact contradicts this (it needs f_D(4) >= 12 M, and no such lower bound on f_D(4) is known).
- Why previous rounds missed it: Lemma S2 fixes ONE auxiliary point and needs it spread; here BOTH auxiliary points may be heavy. The choice of w0 in Q' (rho_k notin L_w0) gives a cover of C_k by spheres about a1 AND a2; slicing C_k along one class of a2 turns that cover into a column-type cover in a1 (two-centre lemma); HR makes "a2 has few heavy slices" a covering statement.
- Constant 11 is not optimised.

Appendix G. Label lemma and G1 for general n — rounds 10–11

Source file: e1088/round10/05_general_n.md

05. General n (target B)

X rainbow-n-free in R^D, n >= 4 fixed. T = {t_1, ..., t_{n-2}} a rainbow (n-2)-subset, D_T its C(n-2,2) distances. For x not in T: L_x = {|x t_1|, ..., |x t_{n-2}|}; x is T-generic if T u {x} is rainbow.

Lemma 5.1 (label structure). [PROVED]

If x, w are T-generic and L_x n L_w = {}, then |xw| in D_T u L_x u L_w. Proof. The n-set T u {x,w} has its C(n,2) - 1 distances other than |xw| pairwise distinct (D_T u L_x and D_T u L_w are rainbow, L_x n L_w = {}), so |xw| repeats one of them. [] (For n = 4 this is round 9 Lemma 3.1(i) and the 5-value condition of 01.)

Lemma 5.2 (bilinear form, linear labels). [PROVED]

Put t_1 = 0 and restrict x to S(0,rho), w to S(0,rho'). Then |x t_i|^2 = rho^2 + |t_i|^2 - 2 is affine in x, and |xw| = d (d in D_T) <=> = (rho^2 + rho'^2 - d^2)/2, |xw| = |w t_i| <=> = (rho^2 - |t_i|^2)/2 + , |xw| = |x t_i| <=> = (rho'^2 - |t_i|^2)/2 . For a fixed column w the T-generic, label-disjoint x therefore lie on C(n-2,2) + n - 1 PARALLEL hyperplanes (normal w; this includes |xw| = rho = |x t_1|) plus n - 3 hyperplanes with normals w - t_i (i >= 2): a non-parallel k-partition with k = C(n-2,2) + 2n - 4. For n = 4: 4 + 1, as in 01.

Lemma 5.3 (embedding, no free lunch). [sketch, not checked numerically]

The construction of 02 extends: give each t_i (i >= 2) its own pair of coordinates carrying the x-label and the w-label, so every abstract instance "rank r, C(n-2,2)+1 global values, n-3 personal row labels and n-3 personal column labels" is realised in dimension r + O(n) with all n-sets T u {x,w} rainbow-free. So again only other n-sets can help.

Theorem 5.4 (conditional analogue of 04, by induction on n). [PROVED as a sketch; constants not optimised]

Assume NP_k(L) with k = C(n-2,2) + 2n - 4 and L(m) = o(m), and assume f_D(n-1) = 2^{o(D)}. Then either |X| <= 2^{o(D)} s'_n(D/2) or X is heavy-dominated: a 2^{-o(D)} fraction of the (n-2)-sets T have a point t in T and a sphere S(t,r) containing 2^{-o(D)} |X| points, which absorbs almost all T-generic points. Sketch. (1) Colour values randomly; x goes to P if L_x is red, w to Q if L_w is blue; a label-disjoint pair lands in P x Q with probability 4^{-(n-2)}. Lemma 5.1 covers P, for each w in Q, by spheres S(w,d) (d in D_T u L_w) and bisectors H(t_i, w): k lifted hyperplanes. NP_k and the type analysis of 04 (Theorem 4.1; types: sphere about w, or bisector H(t_i,w) with t_i fixed) give an orthogonal rectangle, hence the analogue of Theorem 4.1 with 16 replaced by 4^{n-2}. (2) The degree argument of Corollary 4.2 works

with the (n-2)-uniform "label hypergraph": ordered pairs sharing a label number <= sum_v deg(v)^2 <= maxdeg * (n-2)|Y|, so if no sphere about a point of T carries >= |Y|/(4(n-2)^2) of Y, at least half of all pairs are label-disjoint. (3) Lemma 4.4 is replaced by sampling against rainbow-(n-1)-freeness: if the density of rainbow (n-1)-sets is tau, a random m-set with m ~ ((n-1)!/(2 tau))^{1/(n-2)} has few of them, so m/2 < f_D(n-1); with f_D(n-1) = 2^{o(D)} this gives tau >= 2^{-o(D)}, and averaging over (n-2)-sets T gives a T with 2^{-o(D)} |X| generic points. [] So for every n the open residue is the same as for n = 4: (O1) the bipartite label rectangle (non-parallel k-partitions, open), (O2) heavy spheres / multi-piece concentration, (O3) class count.

(C) complete proof

Not attempted in final form: no complete argument exists even for n = 4 (see 04, "Conditional decomposition"). complete_proof.md is therefore NOT written; writing one would misrepresent the state of the problem.

Why the n = 4 endgame (06) does not transfer verbatim [HEURISTIC]

06 uses that a typical point has exactly ONE big class (type-A points are few by Prop 4.7) and then a 1-label kill polynomial. For n >= 6, 2-label hierarchies (round 9, 02) are rainbow-6-free and have points with two (or more) big classes legitimately, so the analogue of Prop 4.7 must allow up to k(n) big classes and the endgame must use k(n)-label kill polynomials (Lemma H, degree k(n)); the bipartite step (Theorem 5.4 (1)) and the heaviness step (5.4 (3)) go through by induction. Whether the multi-class analogues of Steps 4-6 of 06 hold is NOT checked.

Source file: e1088/round11/02_general_n.md

02. General n after Theorem LR: what is now proved, what is still missing

STATUS: general n >= 5 remains OPEN. Below: PROVED pieces (with LR) and the precise missing statements.

Setting: X rainbow-n-free in R^D, s'*n(j) = largest rainbow-n-free set of affine dimension <= j, f_D(n) = s'_n(D) + 1. T = {t_1..t*{n-2}} a rainbow (n-2)-set, D_T its distances, L_x = (|x t_i|)_i, Y_T = T-generic points.

PROVED now (unconditional, given the n-1 case for (G2))

(G1) Bipartite step for all n [PROVED, LR-4 of 01]. If P, Q in Y_T and L_x n L_w = {} for x in P, w in Q, then min(|P|,|Q|) <= 2^{L_k(D+2)} (s'_n(floor(D/2)) + 1) =: mu_n, k = C(n-2,2) + 2(n-2) (LR's single-type conclusion means no further splitting into type classes is needed). Proof: round10/05 Lemma 5.1 makes |xw|^2 a row/column-labelled rank-(D+2) matrix with k types; LR gives a single-type rectangle; each type gives (OR). Consequently (random colouring with 2^{-2(n-2)} loss) |{label-disjoint ordered pairs in Y}| <= 4^{n-2} mu_n |Y| for Y in Y_T, and the degree form: if every value has degree <= |Y|/(4(n-2)) in Y then |Y| <= O(4^{n-2} mu_n). (G2) Heaviness [PROVED assuming f_D(n-1) <= F := 2^{o(D)}]. Any m-set with m >= F contains a rainbow (n-1)-set, so a set Z with |Z| >= F has rainbow-(n-1)-set density >= 1/C(F, n-1) =: tau_n = 2^{-o(D)}. Hence, as in Step 2 of n4_full_proof.md (with Lemma 4.4 replaced by this density bound and pairs (o,a) replaced by T), all but gamma N points of X centre a class of size >= beta N with beta = tau_n gamma/poly(n) = 2^{-o(D)}, unless N <= 2^{o(D)} mu_n. For n = 5 this uses only the n = 4 theorem: tau_5 >= 2^{-O(sqrt D log D)}. So, by induction on n, round10/05 Theorem 5.4 is now UNCONDITIONAL: either |X| <= 2^{o(D)} s'_n(D/2) or X is heavy-dominated with heaviness 2^{-o(D)}.

What the n = 4 endgame used that has no general-n proof yet [OPEN]

(E1) Few big classes per typical point. n = 4: Prop 4.7 (two classes >= M force all but mu3 points into them) and Prop 4.8 (few such centres) leave a UNIQUE big class lambda(o). For n >= 6 this is false (2-label hierarchies are rainbow-6-free and have two robust big classes at every point); for n = 5 we neither have a proof nor a counterexample. Needed: a bound K(n) (or K = o(D/log D)) on the number of classes of size >= M at typical points, with M = 2^{o(D)} s'_n(D/2). (E2) Killing. With Lambda(o) = big radii of o, |Lambda(o)| <= K, Lemma H gives |I| <= C(D+1+K, K)^2 for any I in which every pair {o,a} has |oa| in Lambda(o) u Lambda(a). Needed: the pairs with |oa| outside Lambda(o) u Lambda(a) form a graph of average degree 2^{o(D)} s'_n(D/2). For n = 4 this was Steps 4-6 (bisector charging and equal-label 2-listable rectangles), which use 4-sets {o,a,x,w} through exactly two fixed points. For general n the analogue must use (n-2)-sets T containing o and a and n-4 further points; the Y**-type sets become cells of an (n-2)-dimensional grid of labels, and the analogue of Step 4(b) (one row x one column) becomes a family of LR-instances indexed by which coordinates of L_x are 'big' radii. The LR side is fine (any

bounded number of row/column types); the combinatorial bookkeeping (which cells are small, which pairs are charged where) is not done. HEURISTIC expectation: it goes through for n = 5, 6 with K = n - 3, but this is NOT checked. (E3) Small-class mass. Step 4(c) of n = 4 needs that the degenerate loci (E(o,a) = H(o,a) u S(o,c0) u S(a,c0)) carry only small classes; for general n the degenerate loci of T are C(n-2,2) + (n-2)(n-3)/... spheres and bisectors, and the same "unique big class" information is needed for each t_i.

Assessment

The bipartite obstacle (O1) of rounds 9-10 is gone for every n (LR). What remains for n >= 5 is the class-count / multi-piece obstacle (O2)-(O3) in the endgame, i.e. (E1)-(E2) above. A plausible route: induction on n with the statement "typical points have <= n - 3 big classes and almost all pairs are killed by the big radii of one endpoint", using LR for every bipartite sub-step and Lemma H with degree n - 3. This is a programme, not a proof.

Appendix I. Full n = 4 proof (orthogonal split OR, halving recursion) — round 11

Source file: e1088/round11/n4_full_proof.md

f_d(4) = 2^{O(sqrt(d) log d)}: assembled proof (round 11)

STATUS: every step below is PROVED or CITED (classical results only). The only ingredient that was missing in round 10, Hypothesis LRQ, is now Theorem LR (file 01). Adversarial re-verification: see the end of this file and 00_index.md. Nothing here has been published or externally refereed; treat as a candidate proof.

Theorem

There is an absolute constant C such that every set of more than 2^{C sqrt(d) log2(d+2)} points in R^d contains 4 points whose 6 pairwise distances are distinct. Hence f_d(4) = 2^{O(sqrt d log d)} = 2^{o(d)}.

0. Notation

rainbow 4-set: 4 points with 6 distinct distances. s'(j): the largest size of a rainbow-4-free set of affine dimension <= j (s'(0) = 1; s' is nondecreasing; f_d(4) = s'(d) + 1). Fix X rainbow-4-free, affine dimension <= D, N = |X|, s' := s'(floor(D/2)). S(c,t) sphere, H(p,q) perpendicular bisector. For c in X, a CLASS of c is a nonempty X n S(c,t), t > 0. For o != a in X: rho(x) = |ox|, alpha(x) = |ax|, c0 = |oa|, rad(x) = {rho(x), alpha(x)}, Y(o,a) = {x in X : {o,a,x} has 3 distinct side lengths}. For Y in Y(o,a): RD(Y) = ordered pairs (x,w) in Y^2 with rad(x) n rad(w) = {}; deg_Y(v) = #{x in Y : v in rad(x)}.

1. Tools

(LR) Theorem LR (file 01, PROVED): rank-r matrix, every entry equal to one of k row/column type functions => single-type rectangle of relative size 2^{-L_k(r)} each side, L_k(r) = O(k^3 sqrt(r+2) log(r+2)). (RL) RQ-list-G (special case of LR, all types column functions; PROVED round 9 and again by LR). P, Q in R^D finite disjoint, every w in Q sees P in <= k distances => R in P, C in Q, |R| >= 2^{-L_k}|P|, |C| >= 2^{-L_k}|Q| (L_k := L_k(D+2)), every w in C equidistant from R. Same with rows and columns swapped. [[|pw|^2] has rank <= D+2.] (GL) Cleaning lemma (Gavinsky-Lovett; proof in round9/01). rank <= R', <= 1/(16R') fraction of nonzeros => a zero sub-rectangle with half the rows and half the columns. (OR) Orthogonal split [PROVED]. If R, C in X and = 0 for all x,x' in R, w,w' in C, the direction spaces of aff R and aff C are orthogonal subspaces of R^D, so min(dim aff R, dim aff C) <= floor(D/2) and min(|R|,|C|) <= s'. Each of the following gives it: every w in C equidistant from R (|xw|^2 - |x'w|^2 = |x|^2 - |x'|^2 - 2 independent of w); or R in H(p,w) for all w in C, p fixed (= 0). (H) Lemma H, symmetric form [PROVED, round 8 01: rank of F o F^T]. If F_x lies in a space of functions of dimension m, F_x(x) != 0, and every pair x != y has F_x(y) = 0 or F_y(x) = 0, then the number of points <= m^2. (B) Blokhuis [CITED: A. Blokhuis, Few-distance sets, CWI Tract 7, 1984]: an isosceles set (every 3 points span an isosceles triangle) in R^D has at most C(D+2,2) points. (HP) rank(A o B) <= rank A * rank B for the entrywise product [PROVED, elementary]. (CW) Caro-Wei [CITED, classical]: a graph with

n vertices and average degree dbar has an independent set of size >= n/(dbar+1). (DIM) dim of polynomials of degree <= k in the lifted coordinates (|y|^2, y_1..y_D): <= C(D+1+k, k); degree <= k in y_1..y_D: C(D+k, k). Sphere/bisector equations have degree 1 in the lifted coordinates.

2. Step 1: radius-disjoint pairs are few [PROVED; uses LR]

mu := 3 * 2^{L_5(D+2)} (s' + 1). Claim: |RD(Y)| <= 16 mu |Y| for every Y in Y(o,a). Proof. Colour the reals red/blue independently and uniformly. P = {x in Y : rho(x), alpha(x) red}, Q = {w in Y : rho(w), alpha(w) blue}; P n Q = {}. A pair of RD(Y) involves 4 distinct values, so it lies in P x Q with probability 1/16: fix a colouring with |P||Q| >= |RD(Y)|/16, and assume P, Q nonempty. For x in P, w in Q the five numbers c0, rho(x), alpha(x), rho(w), alpha(w) are distinct (rainbow triangles, disjoint radius sets), so rainbow-4-freeness of {o,a,x,w} forces |xw| to equal one of them. Thus M[x,w] = |xw|^2 (rank <= D+2) is covered by the k = 5 types c0^2 (constant), rho(w)^2, alpha(w)^2 (column functions), rho(x)^2, alpha(x)^2 (row functions). LR gives R in P, C in Q, |R| >= 2^{-L}|P|, |C| >= 2^{-L}|Q| (L = L_5(D+2)) and one type on R x C. Column types / constant: every w in C equidistant from R. Type rho(x)^2: R in H(o,w) for all w in C. Type alpha(x)^2: R in H(a,w). In all cases (OR): min(|R|,|C|) <= s', hence min(|P|,|Q|) <= 2^L s' <= mu and |RD(Y)|/16 <= |P||Q| <= mu max(|P|,|Q|) <= mu |Y|. [] Degree form. If deg_Y(v) <= |Y|/4 for every v, then |Y| <= 32 mu + 2. [Ordered pairs (x,w), x != w, sharing a value: <= sum_v deg(v)^2 <= maxdeg * sum_v deg(v) = maxdeg * 2|Y| <= |Y|^2/2; so |RD(Y)| >= |Y|^2/2 - |Y|, and |Y|^2/2 - |Y| <= 16 mu |Y|.] (Remark, unconditional alternative: round9/03 Lemma 3.1(iii) (via (H), nu = C(D+4,3)^2) gives |Y| <= 2 nu maxdeg_Y.)

3. Step 2: almost every point centres a heavy sphere [PROVED]

tau0 := (1 + 1.2 C(D+2,2))^{-2}, eps1 := 1/(64(D+2)), gamma := eps1/4, beta := tau0 gamma/20. Lemma 4.4 [PROVED from (B)]: a set Z in R^D with |Z| > 1.2 C(D+2,2) has >= tau0 C(|Z|,3) non-isosceles triples. [If the fraction tau < tau0: tau = 0 contradicts (B); else take m = min(|Z|, floor(tau^{-1/2})) > 1.2 C(D+2,2), a random m-subset has <= tau m^3/6 <= m/6 non-isosceles triples in expectation; deleting a point of each leaves an isosceles set of > C(D+2,2) points.] Light point: no class of size >= beta N. Claim: fewer than gamma N points are light, if (T2) N > (64 mu + 4)/(tau0 gamma) and gamma N > 1.2 C(D+2,2) + 4. Proof. Let Lt (light points) have >= gamma N points. Summing over the non-isosceles triples of Lt, some pair o, a in Lt has |Y(o,a)| >= |Y(o,a) n Lt| >= tau0 (|Lt| - 2) >= tau0 gamma N/2 =: y0. In Y = Y(o,a), deg_Y(v) <= |X n S(o,v)| + |X n S(a,v)| < 2 beta N = 0.1 tau0 gamma N <= |Y|/4, so |Y| <= 32 mu + 2 < y0. []

4. Step 3: typical points have exactly one big class [PROVED; uses Step 1, (RL), (GL)]

eps := 1/(16 C(D+4,4)), M := max(16 mu/eps^2, 2^{L_4+2}(s'+1)), mu3 := 4 * 2^{L_4}(s'+1), mu4(k) := 4 * 2^{L_k}(s'+1). Assume (T3): beta N >= M, N >= 32 C(D+3,2)(mu3 + 3), N >= 192(D+2), N > 4 * 2^{L_2}(s'+1), N > 4 * 2^{L_1}(s'+1), gamma N + mu4(1) + mu4(2) <= N/2. Prop 4.6. K_i, K_j classes of o (radii rho_i != rho_j), a in X \ (K_i u K_j u {o}), K_i' = {x in K_i : |xa| not in {c0, rho_i, rho_j}}, likewise K_j'. Then sum_t |K_i' n S(a,t)| |K_j' n S(a,t)| >= |K_i'||K_j'| - 8 mu (|K_i'| + |K_j'|). [K_i' u K_j' is in Y(o,a); (x,w) in K_i' x K_j' is radius-disjoint iff |xa| != |wa|; RD counts ordered pairs; Step 1.] Consequence: if |K_i'|, |K_j'| >= 16 mu/eps, one sphere S(a,t*) carries >= (1-eps)|K_i'| and >= (1-2eps)|K_j'|. Prop 4.7. If o has two classes K_i, K_j of size >= M, then |X \ (K_i u K_j u {o})| <= mu3. [For a in A := X \ (K_i u K_j u {o}): if |K_i'| < 16mu/eps <= eps M, a sees K_i at the 3 distances c0,*

rho_i, rho_j up to eps|K_i| points; similarly for K_j; if both are >= 16mu/eps, Prop 4.6 gives 4 distances (with t) up to eps|K_i| points of K_i. So A = A_i u A_j (a in A_i sees K_i at <= 4 distances up to eps|K_i| exceptions). Say |A_i| >= |A|/2. B[x,a] = prod_{s<=4}(|x-a|^2 - r_{a,s}^2) (x in K_i, a in A_i) is a degree-4 polynomial in x on the sphere K_i (o = 0: |x-a|^2 = rho_i^2 + |a|^2 - 2), rank <= C(D+4,4), density <= eps. (GL): K'' x A'' (halves) with B = 0; (RL) with k = 4; (OR); |R| >= 2^{-L_4} M/2 > s', so |A''| <= 2^{L_4} s' and |A| <= mu3.] Prop 4.8. If all but e points of X lie on <= k spheres centred at h, for every h in Co, and N >= 32 C(D+1+k,k)(e+2), N > 4 * 2^{L_k}(s'+1), then |Co| <= mu4(k). [H' = half of Co, X1 = X \ H'; in [|xh|^2] (x in X1, h in H') each column takes its <= k sphere values except <= e entries; exceptions = support of prod_s (|x-h|^2 - r_{h,s}^2), degree k in lifted x, rank <= C(D+1+k,k), density <= 2e/N; (GL), then (RL) and (OR) with |R| >= 2^{-L_k} N/4 > s'.] Typical points. Type A = heavy points with two classes >= M: by Prop 4.7 each has all but mu3 + 1 points on 2 spheres about it, so by Prop 4.8 (k = 2, e = mu3 + 1) there are <= mu4(2) of them. Points with all but eps1 N + 1 points on one sphere about them: <= mu4(1) (Prop 4.8, k = 1; 32(D+2)(eps1 N + 3) <= N by N >= 192(D+2)). T := heavy points of neither kind; |T| >= N - gamma N - mu4(1) - mu4(2) >= N/2. For o in T: a unique class K_1(o) of size >= M (it has >= beta N points), radius lambda(o); every other class has < M points; Z_o := X \ (K_1(o) u {o}) has > eps1 N points.

5. Step 4: pairs with different labels [PROVED; uses Step 1]

o, a in T, lambda(o) != lambda(a), c0 not in {lambda(o), lambda(a)}. Then Z_o or Z_a has all but m3 := max(8M, 32 mu + 2) + 3M + 16 mu + 1 of its points in H(o,a). Proof. rho1 = lambda(o), t1 = lambda(a), Y = Y(o,a). (a) Y = **{x in Y: rho(x) != rho1, alpha(x) != t1}: a value v is carried there by <= |S(o,v) n X| points with rho = v (v != rho1: < M) and <= |S(a,v) n X| with alpha = v (v != t1: < M); so deg < 2M and the degree form gives |Y|** <= max(8M, 32 mu + 2). (b) Y1 = *{rho = rho1, alpha != t1}, Y*1 = {rho != rho1, alpha = t1}, p, q their sizes. For x in Y1*, w in Y*1 the radius sets meet iff alpha(x) = rho(w); rho(w) != t1 (w in Y), so S(a, rho(w)) is a small class: each w has < M such x. Step 1 on Y1 *u Y*1 (cross pairs counted in both orders): 2(pq - Mq) <= 16 mu (p+q), whence min(p,q) <= M + 16 mu. (c) If q <= M + 16 mu: Z_o n Y lies in Y* *u Y*1; Z_o \ Y lies in {a} u H(o,a) u S(o,c0) u S(a,c0), the two spheres being small classes (c0 != rho1, t1). So |Z_o \ H(o,a)| <= m3. If p <= M + 16 mu: same for Z_a. []

6. Step 5: charging [PROVED; uses (GL), (OR)]

Assume (T5): eps1 N >= 16(D+1) m3 and eps1 N > 2s'. For o in T, A(o) := {a != o : |Z_o \ H(o,a)| <= m3} has |A(o)| <= 2s' + 1. [F[x,a] = |xa|^2 - |xo|^2 = |a|^2 - |o|^2 - 2 on Z_o x A(o): rank <= D+1, zero iff x in H(o,a), density <= m3/|Z_o| <= 1/(16(D+1)). (GL): Z'' x A'' with = 0, i.e. (OR) applies; |Z''| >= |Z_o|/2 > s', so |A''| <= s'. (If a in Z_o, the entry (a,a) = -c0^2 is one of the exceptions, so the zero block has disjoint row and column sets.)] Hence the number of pairs {o,a} in T as in Step 4 is <= (2s'+1)|T| (each has a in A(o) or o in A(a)).

7. Step 6: pairs with equal labels [PROVED; uses Step 1, (GL), (RL)]

(a) o, a in T, lambda(o) = lambda(a) = lambda, c0 != lambda: all x in Z_o but m3' := max(8M, 32mu+2) + 2M + 1 satisfy |xa| in {|xo|, lambda}. [Y **as in Step 4 (both big classes have radius lambda, all other classes small); Z_o n Y \ Y** has alpha = lambda; Z_o \ Y in {a} u H(o,a) u two small classes.] (b) Assume (T6): eps1 N >= 16(D+2)^2 m3' and eps1 N > 2^{L_2+1}(s'+1). Then B(o) := {a in T \ {o} : lambda(a) = lambda(o), |oa| != lambda(o)} (a subset of Z_o) has |B(o)| <= mu7 := 4*2^{L_2}(s'+1). [On Z_o x B(o) each row x of [|xa|^2] takes the two values {rho(x)^2, lambda^2} except on the support of G

= (|xa|^2 - |xo|^2) o (|xa|^2 - lambda^2), rank <= (D+2)^2 by (HP), density <= m3'/|Z_o| (per column); the diagonal entries (x = a) are nonzero (= c0^2 lambda^2), so (GL) yields a zero block Z'' x B'' with disjoint index sets; (RL) with ROWS 2-listable gives R in Z'', C in B'' with every x in R equidistant from C; (OR) and |R| >= 2^{-L_2}|Z_o|/2 > s' give |C| <= s', so |B''| <= 2^{L_2} s'.]

8. Step 7: one-label endgame [PROVED; uses (H), (CW)]

F_o(y) := |y - o|^2 - lambda(o)^2 (degree 1 in lifted coordinates, space of dimension D+2), F_o(o) != 0. A pair {o,a} in T with |oa| in {lambda(o), lambda(a)} is killed from one side. The remaining pairs are those of Step 4 (<= (2s'+1)|T| in total) or of Step 6 (<= mu7 per point). So the graph of unkilled pairs on T has average degree <= dbar := 4s' + 2 + mu7; (CW) gives an independent set I with |I| >= |T|/(dbar+1), all of whose pairs are killed; (H): |I| <= (D+2)^2. Therefore N <= 2|T| <= 2 (D+2)^2 (4 s' + 3 + mu7) unless one of (T2), (T3), (T5), (T6) fails.

9. The recursion

Conventions: all "halves" from (GL) are rounded down (so |A(o)| <= 2|A''| + 1, |B(o)| <= 2|B''| + 1, |Co| <= 2|H'| + 1); mu3, mu4(k), mu7, 2s'+1 absorb this. In Prop 4.8 with k = 1 the exception bound e = eps1 N + 1 depends on N, but the hypothesis 32(D+2)(e+2) <= N is implied by N >= 192(D+2) and the conclusion mu4(1) does not. Every threshold in (T2)-(T6) is of the form poly(D) * 2^{O(sqrt D log D)} (s'+1): mu, M, mu3, mu4(k), m3, m3', mu7 are all <= poly(D) 2^{L_5(D+2) + L_4(D+2)} (s'+1) and 1/beta, 1/(tau0 gamma), 1/eps1 are poly(D). Hence for all D >= 1 s'(D) <= A(D) (s'(floor(D/2)) + 1), A(D) = 2^{C1 sqrt(D) log2(D+2)}, Since s' >= 1, log2(s'+1) <= log2 s' + 1; iterating over D, floor(D/2), ..., 1, 0 (at most log2 D + 1 levels, s'(0) = 1): log2 s'(D) <= sum_{0 <= i <= log2 D} (C1 sqrt(D/2^i) log2(D+2) + 1) = O(sqrt D log D). So f_d(4) = s'(d) + 1 = 2^{O(sqrt d log d)}. []

10. Adversarial re-verification (this round)

- Dependency audit: Step 1 is the ONLY place where round 10 used a hypothesis; it now uses LR (k = 5, rank D+2). (RL) = LR with column types only (and its transpose). (GL), (H), (OR), Lemma 4.4, Props 4.6-4.8 re-derived above.
- Each application of a rectangle lemma ends in (OR) with BOTH sides rainbow-4-free subsets of X (true: subsets).
- Each (GL) application: the matrix's support is exactly the exception set (sphere / bisector equations; Hadamard products of such), the stated rank bound, density hypothesis implied by the thresholds; diagonal coincidences (x = a) are nonzero entries, so zero blocks have disjoint index sets.
- No circularity: thresholds depend only on D and s'(floor(D/2)); LR's loss depends only on r = D+2 and k.
- Step 4(c) count: m3 = |Y**| + (M + 16mu) + 2M (two small classes) + 1 (the point a or o), as in round 10.
- Sanity: replacing LR by the trivial 2^{-O(r)} bound returns s'(D) <= 2^{O(D)} s'(D/2), consistent with the classical 2^{O(d)} (Erdos-Straus); the new bound 2^{O(sqrt d log d)} is far above the known lower bounds (f_d(4) is at least polynomial in d; no superpolynomial lower bound is known to us), so no contradiction.
- Two independent referee passes (2026-10-19): Theorem LR "correct with minor fixes" (all cosmetic, applied in 01: degenerate NF3 cases, constants in pair rule, K = 256(k-1), rounding, empty minority); this file "correct with fixes" (all cosmetic, applied: a != o in B(o), the finite sum in Section 9, (HP) listed, rounding conventions).

Appendix J. Lemma H (one-sided killing) — round 8

Source file: e1088/01_hadamard_bipeeling.md

01. Hadamard-product ("bi-peeling") lemma and label sets [session jz, 2026-10-18]

Prior work cited: peer notes earlier in-project notes (not public) (rounds 1-14) and erdos1088_partial.tex (L1 few-distance lemma, L2 peeling lemma, THM-R, THM-R2, twin lemmas).

Notation. For x in R^d write u(y) = (|y|^2, y_1,...,y_d) ("lifted coordinates"). P_k := polynomials of degree <= k in the d+1 lifted coordinates, restricted to R^d; dim P_k <= C(d+1+k, k). Every sphere/hyperplane equation |y-a|^2 - r^2, - c lies in P_1.

Lemma H (bi-peeling). PROVED

Let X be finite with a linear order <. For each x in X let F_x, G_x be real functions on X with F_x(x) G_x(x) != 0, and suppose for all x < y : F_y(x) * G_x(y) = 0 (i.e. the later point's F kills the earlier point, or the earlier point's G kills the later one). If all F_x lie in a vector space of functions of dimension m_F and all G_x in one of dimension m_G, then |X| <= m_F * m_G .

Proof. A_{y,x} := F_y(x) has rank <= m_F (its rows are restrictions of elements of an m_F-dim space). B_{x,y} := G_x(y) has rank <= m_G. H := A o B^T (entrywise product), H_{y,x} = F_y(x) G_x(y). Rows/columns ordered by <: H_{y,x} = 0 for x < y, so H is upper triangular with nonzero diagonal, rank H = |X|. And rank(A o B^T) <= rank(A) rank(B) (A o C = sum_{i,j} (a_i o c_j)(b_i o d_j)^T for A = sum a_i b_i^T, C = sum c_j d_j^T). []

Special cases. * G_x == 1 (or F == 1): the ordinary peeling lemma (L2 of the peer), with m = C(d+D,D). * F = G: symmetric condition "for every pair x != y: F_x(y) = 0 or F_y(x) = 0" (no order needed), bound |X| <= m^2. ("every pair is killed from at least one side")

What is new compared to L2: a pair only has to be killed from ONE side, chosen per pair. The price is squaring the dimension, which is harmless for both targets poly(d) and 2^{o(d)}.

Corollary L (label sets are polynomial). PROVED

Call X a k-label set if there are lists L(x) of <= k positive reals with |xy| in L(x) u L(y) for all x != y. (k-label sets are rainbow-n-free for n >= 2k+2; for k=1, rainbow-4-free.) Then |X| <= C(d+1+k, k)^2 ; in particular a 1-label set has |X| <= (d+2)^2. Proof. F_x(y) = prod_{r in L(x)} (|y-x|^2 - r^2) in P_k, F_x(x) = prod(-r^2) != 0, and for x != y, |xy| in L(x) gives F_x(y)=0, |xy| in L(y) gives F_y(x)=0. Apply Lemma H with F=G. []

Consequence for the problem: the peer's Round-5 worry ("if k-label sets could be exponentially large in d the answer would be NO") is settled: k-label sets have at most d^{O(k)} points; even k = eps*d labels per point gives |X| <= C(d+eps d, eps d)^2 = 2^{c(eps) d}, c(eps) -> 0.

Corollary L' (mixed certificates). PROVED

Suppose each x has a polynomial F_x in P_D with $F_x(x) \neq 0$ and every pair $\{x,y\}$ has $F_x(y)=0$ or $F_y(x)=0$. Then $|X| \leq C(d+1+D, D)^2$. So for $2^{o(d)}$ it suffices to find, for each point, a "kill polynomial" of degree $o(d / 1)$ such that every pair is killed from at least one side ("D-bi-peelable"); for poly(d), D = O(1).

Remark (what kill polynomials can do cheaply, degree in P_1 units): - spheres around x with radii in a list L(x) (kills x's distance classes), - any sphere/hyperplane not through x, - the peer's twin polynomial (17 quadric/linear factors) for a rainbow triangle (x;y,z), - a single point $w \neq x$ (one hyperplane through w avoiding x). In the one-sided peeling the obstruction was lex chains $w_1,\ldots,w_k,x$ ($|w_i w_j| = |w_i x|$, $i<j$): x cannot kill them cheaply. In bi-peeling each w_i kills x and all later w_j by ONE sphere $S(w_i, |w_i x|)$ -- provided w_i "chooses" that radius. So chains become a LABEL problem; see 02_*.md.

Appendix K. Independent referee report on the general-n proof (referee 15)

Source file: e1088/referee15/REFEREE_REPORT.md

Referee report (referee15) on round14: general-n proof that f_d(n) = 2^{O_n(sqrt d log d)}

Referee: independent agent; I wrote none of the material under review. I re-derived every step of 01–05 by hand (working notes: NOTES.md). Independent code: code/ref_tests.py, ref_tests2.py, ref_tests3.py, logs alongside. I also checked that general_n_full_proof.md contains 01–05 verbatim (0 missing lines) plus a statement and a dependency table that agree with them.

Verdict

item	verdict
(1) Thm 1.3 (single-cap R+)	**Faithful translation.** Constants and quantifiers carry over. It is not written out, which is an expository gap (fix X1).
(2) Lemma 1.1, Thm 1.4, Lemma 1.2, Cor 1.5	**Correct.** There is no circularity, and caps and ratios behave as claimed.
(3) Core Lemma (02)	**Correct.** F1–F5 are maintained, validity is a constant-type condition, the S-coincidence costs are right, and bisectors are handled correctly.
(4) Slicing Lemma (03), E1_r, K_r	**Correct.**
(5) Threshold tower	**Correct.** The order of definition is acyclic, Gamma_r = (D+2)^{c_n}, sigma_min = 2^{O_n(sqrt D log D)}(s'+1), and the halving recursion closes.
(6) Tests	About 340k independent exact checks on genuinely rainbow-n-free sets (n = 6, 7) found 0 violations. Threshold-dependent claims cannot be tested numerically.
Whole claim	**CORRECT WITH FIXABLE (EXPOSITORY) GAPS, conditional on the in-project dependencies.** I found no mathematical error in round 14.

The conclusion f_d(n) = 2^{O_n(sqrt d log d)} is exactly as reliable as its in-project dependencies. These are, in order of risk: **Theorem LR** (round11/01), then **R+ Step A** (round12/03 with fix F1), then **G1** (which is LR plus a 2-colouring). None has been checked outside this project, and LR claims more than the published literature (see "Weakest links"). The work needs a human expert check before it is described as a solution of Erdős #1088.

(1) Theorem 1.3: the single-cap restatement of R+

I checked R+ as written (round12/03) together with REFEREE_R fixes F1–F3. R+ reads its hypotheses about W_S in exactly three places.

- **Base (level n-2).** The degree form of G1 needs deg_Y(v) = sum_s |Y n S(s,v)| < (n-2) * (cap) <= |Y|/(4(n-2)). With a single cap sigma this gives |W| <= max(4(n-2)^2 sigma, $2^{4^{n-2}} mu + 4$) =

4(n-2)^2 sigma, once sigma >= 24^{n-2} mu + 4 (this is in sigma_+). So Gamma_{n-2} = 4(n-2)^2 is right. I re-derived the degree-form arithmetic: (3/4)|Y|^2 - |Y| <= 4^{n-2} mu |Y|.

- **Case B (|Big_b(Q sigma)| <= 1).** W' = {x in W{b} : S u {b,x} rainbow, |xb| notin Big_b} with cap Q sigma is a level-(j+1) configuration: the classes of s are < sigma <= Q sigma and the classes of b are < Q sigma. Claim(j+1) applies because Q sigma >= sigma_min. The remaining unkilled points lie on the r^2 spheres S(s,|bs'|), which meet W in < sigma points each.
- **Step A (case A).**
- S' is any n-4 points of S. The spread ratio is (cap)/|C_i| <= sigma/(Q sigma) = 1/Q, which is the same as tau_{r-1}/tau_r in R+.
- In the F1 branch, theta_r = Q^{-1/2} tau_r = Q^{1/2} sigma. The S'-coincidence cost is n(n-4) sigma (1/|P| + 1/|Q|) <= 2n^2 Q^{-1/2}.
- The G1 cost 2*4^{n-2} mu Q^{1/2}/tau_r is small because tau_r = Q sigma >= sigma_min >= tau_min^{R+}.
- A_0 lies inside W \ {b} (F2). The exclusions S(s,rho_i), S(s,rho_j) for s in S' cost 2(n-4) sigma, and S(s,|bs'|) costs r^2 sigma.
- I re-checked that T = {b,a} u S' is rainbow and that rho_i, rho_j notin D_T for a in A_0.
- **Recursion.** These bounds give Gamma_j = P_j^+ (2 Q Gamma_{j+1} + 2(j^2+2n) + 1), using e_A <= sigma <= Q Gamma_{j+1} sigma and 1 <= sigma. The degree is e = 2 + 2j + C(j,2), which gives P_j^+ = C(D+1+e, e)^2 as stated. F3 (tau_{-1}) is moot, because S' is nonempty for every j >= n-4 >= 1.
- **Quantifiers.** R+ never uses points outside the current ground set, so "for all S built from X" becomes "for all configurations (S, W, sigma) with sigma >= sigma_min". Gamma_j does not depend on sigma.

(2) The induction framework

- **Lemma 1.1.** The list of non-rainbow events for S u {c,x} is complete. I also confirmed this by exact test on actual configurations: the derived (S u {c}, V_1, theta) passes a full configuration check every time.
- The cap for s is inherited from W.
- The cap for c holds because every class of c meeting V_1 meets V in < theta points.
- theta >= sigma >= sigma_min, and Gamma'*r* = *Gamma*{r+1} + r^2.
- **Lemma 1.2.** For each case, F_b(x) != 0 means either x lies on one of the r^2 small spheres, or x is in W'. Here (S u {b}, W', tau) is a configuration, since a radius notin R_b u Big_b gives a class < tau. Lemma H (A o A^T is diagonal with nonzero diagonal, so |I| <= m^2) and Caro–Wei give the bound P (2 Gamma_{r+1} tau + 2r^2 sigma + 1). This holds for every tau >= sigma.
- **No circularity.** The chain is R+ (levels >= n-4), then for each r: HR_r uses only Claim(r+1); E1_r uses HR_r at the same level and the same configuration; Claim(r) uses E1_r and Claim(r+1). Every Claim(j) is quantified over all configurations and all caps >= sigma_min. So applying Claim(j+1) with theta >= sigma (Lemma 1.1, Step 3, slicing (a)) or with tau = ratio_r sigma (Lemma 1.2) is legitimate.
- Every absolute error term is absorbed by a lower bound on sigma_min. These terms are e, e_A, 2*4^{n-2} mu + 4, mu and 1.
- Gamma_j is therefore a pure ratio. Each application has threshold >= sigma: theta' = theta + r sigma in Step 3, and theta_sl |C_k| >= 2n(n+K) sigma in slicing.

(3) Core Lemma (02)

- **F1–F5.**

- Initial family (Thm 1.4) and slicing (d): I re-derived every cross-coincidence. Removing the spheres S(s, {|cs'|} u {lambda_k'}) is exactly what is needed for (F4) and (F5) of the new family.
- Every condition passes to subfamilies.
- Disjointness follows from beta_1.
- **Validity is a constant-type condition.** I verified (5.1) and the formula for D_T. On V_lm:
- k* is never l or m, since beta(l), beta(m) are in Ty;
- for every k != k*, beta(k) n D_T = {} by F3 and F5. So all pairs in I0 are valid. The counting "g <= |I0| - 4 implies every k' in I0 is Cov" is correct, which gives g + #Cov >= 6.
- **Step 0 and Step 1.** The coincidence list is complete:
- |xs| against beta(l), beta(m) is excluded by F5;
- |xs| against |a_i s'| or |a1a2| gives the r(2r+1) spheres Sm;
- |xs| = |xa_i| gives Hyp.

In Step 1(i), the |xs| in L_w0 cases give r(r+2) further spheres. The bound |W| <= w1 holds because |L_w0| = r + p + 2 = n - 2. The G1 P x Q form follows from the ordered-pair form on Y = P u Q. I reconstructed G1 itself (random red/blue colouring of the reals, then LR-4 and OR on the complete label-disjoint bipartite pair) and it gives mu = 2^L (s'+1). - **Step 2.** The S(a_i, beta(k)) are disjoint across k, so c(|J| - 2) < 2|J| eps |C_k'|. Correct. - **Steps 3 and 4.** - (3.1) holds. The sphere S(s,lambda) is removed from R_lambda, so the points of H(s,a2) are gone. - Per-column density is <= |R_lambda|/(16 (D+2)^{k3}) because |R_lambda| >= theta >= theta_rel |C_j|. - The pigeonhole over w+1 heavy radii is right. - In the few branch, HR is used with R_{a2} contained in Lam(a2). The bound for x in C_j is |C_j n S(a2,lambda)| < theta + r sigma. - Step 4 uses (P4) for the (r+1) sigma part. - Whether a2 is "w-few" depends only on (a2, j, w). This is what Corollary 4.1 needs. - **Costs through S.** Every small-sphere cost is a class of some s in S meeting W, so < sigma each by the cap. I re-checked (P1), (P2), (P4) and (P5), k5 <= k3(w1), w1 + w2 + 2 < n^2, and 1/eps <= 32^{10} Gamma' (D+2)^{k_c}. - *Bisectors. Row type |x a|^2 = |xs|^2 on R x C means R lies in H(s,a) for every a in C. Then (x - x').(s - a) = 0, so dir aff R is orthogonal to dir aff(C u {s}), and OR gives min(|R|,|C|) <= s'. The matrix for each factor has rank <= D+1, so HP holds. LR explicitly allows mixed row, column and constant types. - GL.** Only the covering (low-density) form is used. I re-proved the real-matrix version: Markov on the columns, then a column basis, gives a zero rectangle of size (7/8 of the rows) x (1/2 of the columns). The proof should state this (fix X3).

(4) Slicing Lemma (03)

- **König dichotomy.** For c in Z_bad and k notin A(c), every edge at k ends in Lam0(c). So a point x off Lam0 u Phi(c) u beta(k) and off the bisectors H(c,s) lies in a light class with radius notin R_c. That is exactly HR's premise, with theta = theta_sl |C_k| >= sigma.
- **Bad centres.** Averaging gives |Z_k| >= 3K|Z_bad|/(6K-1) > |Z_bad|/2. Then GL, LR and OR (types: Lam0(c) and |cs| as column functions, D_{S u B}, beta(pi), beta(k) as constants, |xs| as rows; k_sl factors) give |Z_bad| <= 2e. Also |C_pi| >= M_sl > 2e.
- **Label conflicts.** By F3 each lambda_k hits at most one old label. There are no loops, because of the edge rule lambda_k notin beta(k). So out-degree <= 1, the graph has at most 3K edges on 3K vertices, and Caro–Wei gives an independent set of size >= K. Independence rules out lambda_k in beta(k') in both directions, which (F3) needs. The bound is tight (K disjoint 3-cycles); my exhaustive test confirms it.
- **K_r.** K^(p) = 6 K^(p+1) and K^(n-4-r) = 12, so K_r = 12 * 6^{n-5-r}. The m_p recursion and ratio_r = 2 m_1/sigma + 2n(n+K_r) are consistent, and tau/2 >= m_1 after the initial cleaning.

(5) The threshold tower

The constants are defined in this order: Gamma_{n-2..n-4}, then for r = n-5..0 Gamma'*r, ratio_r, Gamma_r, and last sigma_min = max(sigma+*, sigma_abs). Nothing loops back: - ratio_r depends on (n, D, Gamma'*r); - k and L depend on (n, D) only, since k_sl and k_c depend on K, not on Gamma; - sigma_min depends on (n, D, s', mu, Q, L*{k'}, L*).

Growth: - ratio_r <= C_n Gamma'^{n-3-r} (D+2)^{O_n(1)}, and Gamma_r ~ P ratio_r Gamma_{r+1}. So log_{D+2} Gamma_0 is about (n-2)!/2 times the top exponent, plus lower-order terms. This is a constant c_n. - P(K_r - 1, r) = (D+2)^{O(6^n)}, k = *O(6^n), L* = O(6^{3n} sqrt D log D). - sigma_min = 2^{O(L*) + O(L_{k'}) + O_n(log D)} (s'+1).

Halving: s'(D)+1 <= prod_i 2A(D/2^i) = 2^{O_n(sqrt D log D)}. One cosmetic point: Cor 1.5 says sigma_min depends "only on n, D, s', mu, L*", but it also depends on Q and L_{k'} through tau_min^{R+}.

(6) Independent tests

These tests are my own and differ from the author's check_core.py, which samples random grid points that need not come from a rainbow-n-free set. I ran the full pipeline on sets that are genuinely rainbow-n-free, in exact integer arithmetic.

Test sets. - **Orthogonal-sphere sets** X = {0} u U_i Y_i. - Y_i is the set of integer points of x^2+y^2+z^2 = R (R in {3,5,6,9}), placed in orthogonal blocks with |y|^2 = c0 + i. The parameter c0 is varied so that distances of different kinds collide often. - X is rainbow-n-free: two points in Y_i and one point elsewhere give equal cross distances; each Y_i has fewer than C(n,2) distances; and the index set contains no WEAK Sidon (n-1)-subset. That holds for [1..7] when n = 6 and [1..12] when n = 7, which I computed. - My first run used Golomb-ruler bounds ([1..11], [1..17]). Sampling detected rainbow 6-sets, and the label-disjoint-pair test fired. This was my construction bug, not the author's. It is fixed, and it shows the detector works. - **Few-distance lattices** {0,1}^8, {0,1,2}^4, and a mixed set.

Pipeline. For r = 0, 1, 2, I built configurations with the exact minimal cap and checked: - Lemma 1.1, as a configuration check on V_1, plus |V_2| < r^2 sigma; - Lemma 1.2, i.e. that unkilled points lie in W' or on small spheres, and that W' is a configuration; - the Thm 1.4 initial family; - slicing (a)–(d), with real Hopcroft–Karp matchings, König covers checked for size and coverage, the k-few/HR premise, out-degree <= 1, maximum independent set >= |I_1|/3, and global F1–F5 on the new family; - Core Steps 0, 1(i) (exact covering with the actual Ex), 1(ii) (exact count), S1b, Step 2 (exact inequality), (3.1), (5.1), the formula for D_T, |k| <= *1, validity for k != k*, and the |W| <= w1 bound; - for p = n-4-r: label-disjoint generic pairs satisfy |xw| in D_T u L_x u L_w. This is the one implication that really uses rainbow-n-freeness.

Separate checks. - HP ranks with mixed row, column and constant factors; - OR for the row type; - the conflict digraph, exhaustively for 3K = 6 and by sampling for 9 and 12; - the per-column form of GL.

Totals.

run	checks
ref_tests.py, seed 1	140,759
ref_tests.py, seed 2	125,469
ref_tests2.py	20,576
ref_tests3.py	53,043

That is about 340k checks with **0 violations**.

Limits. - The cap, GL density and OR thresholds are huge (sigma_min >= 2^{L*}(s'+1)), so every threshold-dependent conclusion is vacuous on testable sets. Only the inclusions and implications can be tested. - On the few-distance lattices the families die after one or two slicing rounds, because there are too few distinct labels. - The orthogonal-sphere sets allow at most 7 or 12 classes, below the 12 or 72 the Core Lemma needs. So the Core Lemma's contradiction is never exercised; only its steps are.

Fixes (all expository; none changes a statement or a constant's order of magnitude)

- **X1 (Thm 1.3).** Replace the translation remark by a short single-cap proof:
- the base, through G1's degree form with cap sigma;
- case B, through the configuration (S u {b}, W', Q sigma);
- Step A with S' a subset of S of size n-4 and the costs r^2 sigma + 2(n-4) sigma, the F1 branch at Q^{1/2} sigma, and G1's cost needing Q sigma >= tau_min^{R+};
- Lemma S2 and the four-case analysis cited verbatim.

This is the only round-14 item that is "cited" rather than written out, and the induction depends on it. - **X2 (Lemma 1.2).** Say explicitly that P(K,r) = (dimension)^2 comes from Lemma H's squaring, and that the "+1" and the absolute terms are absorbed by sigma >= 1 and sigma >= sigma_min. - **X3 (GL).** State and prove the real-matrix, average-density version used in Core Step 5 and slicing (a). Three lines suffice: Markov on the columns, then a column basis. - **X4 (Core Step 3/Cor 4.1).** Say that "a2 is w-few on C_j" depends only on (a2, j, w), and that Step 3's conclusion is "not w-few implies |Z| <= (w+1)e". - **X5 (Cor 1.5).** List Q and L_{k'} among sigma_min's dependencies, and state c_n ~ (n-2)!/2 times the top exponent explicitly. The "g_0 = n^{O(n)} 6^{O(n)}" in 05 is consistent with this. - **X6 (05).** Note that check_core.py tests implications on arbitrary integer points, not on rainbow-n-free sets. The one step that uses rainbow-n-freeness (the label-disjoint pair typing behind G1) is therefore untested there; my ref_tests.py covers it.

Weakest links (dependencies)

1. **Theorem LR (round11/01).** Every round-14 step uses it: the single-type rectangles in Core Steps 3–5, slicing (a), R+ Step A, and G1 (through mu).
2. It is new and unpublished. It claims 2^{-O(k^3 sqrt r log r)} single-type rectangles for rank-r matrices covered by k row, column or constant types.
3. Its column-only special case (RQ-list) would already improve the bound that Singer–Sudan state as best known for parallel k-partitioned configurations (2^{-O~(d)}).
4. Referee12 judged it correct. The constants-only case is a believable extension of Lovett/Rothvoss (largest-gap normalisation, margin rounding, GL cleaning, induction on the number of types). The row/column-function case rests on Lemmas A, B and NF, which I did not re-referee.
5. If LR fails, the whole chain (n >= 4) falls back to 2^{O(d)}.
6. **R+ Step A (round12/03 + F1).** This is the only place where a density estimate (S2, giving eta^{1/2} and eta^{1/4} losses) is turned into a covering estimate. It went through one AI referee, and a real threshold gap (F1) was found and repaired there.
7. **G1.** It follows from LR by the 2-colouring argument, which I reconstructed. It carries no risk beyond LR.

The round-14 material itself (Lemma 1.1, Lemma 1.2, the Core Lemma, the Slicing Lemma, E1_r and the tower) is, in my judgement, correct. It is a sound generalisation of the round-13 n = 5 argument. The new

ideas hold up: - centres with constant labels in place of spread points; - relative HR at every level from Claim(r+1); - König plus a conflict-free independent set for slicing.

The main claim should still be reported as **conditional on LR (and R+ Step A)**, refereed by AI only, until an external expert has checked LR.

Appendix L. Referee report on E1(5) (referee 14)

Source file: e1088/referee14/REFEREE_REPORT.md

Referee report (referee14) on round13/01_E1_n5.md: Theorem E1(5) and f_d(5) = 2^{O(sqrt d log d)}

Referee: independent agent. I did not write any of the material under review. Every step was re-derived by hand. Working notes: notes1_rederivation.md. Code: code/ref14_tests.py and code/ref14_greedy.py, with logs alongside.

Verdict

item	verdict
Theorem E1(5): every point has <= 11 classes of size >= M	**CORRECT WITH FIXABLE (EXPOSITORY) GAPS**. No mathematical error found.
HR as used (hereditary relative class bound)	**CORRECT.** It follows from R+ levels 1-3 run on an arbitrary ground set V. It is not stated as a lemma anywhere and must be written out (fix A).
Chain E1(5) => Theorem R (K = 11) => halving recursion => f_d(5) = 2^{O(sqrt d log d)}	**CORRECT**, conditional on the cited in-project results: LR, and through it G1, plus R+ Step A after fix F1 of REFEREE_R.
Adversarial examples	No step's threshold-free claim fails: about 150k exact checks, 0 violations. The threshold-dependent conclusions cannot be tested numerically (see section 4).

I tried to break the proof at every point the brief lists and could not. All issues found are presentational. The only substantive risk to the n = 5 conclusion lies upstream: Theorem LR (unpublished, stronger than the published rank/rectangle bounds) and R+ Step A. I re-derived the n = 5 instance of Step A in outline (section 2) and found no gap, but I did not re-referee LR.

1. E1(5), step by step

Notation is as in the file: T = {b, a1, a2} and D_T = {rho_l, rho_m, |a1a2|}. Here mu = mu_5 = 2^{L_9(D+2)}(s'+1), because G1 for n = 5 has k = C(3,2) + 6 = 9 types.

Genericity (1). x in C_k is not in T; validity makes rho_l, rho_m, |a1a2|, rho_k distinct, so T u {x} is non-rainbow iff |xa_i| in D_T u {rho_k} or |xa1| = |xa2|. Correct (and validity is genuinely needed for (1)).

Step 1(i), the covering trick with no spread point. - G1 in count form (round11/02): use a random 2-colouring of the reals, with loss 2^{-6} for n = 5, and the bipartite G1 with 9 types. This bounds the ordered label-disjoint pairs in Y = P u Q (a subset of Y_T) by 64 mu |Y|. - The pairs (w, x) in Q' x P are among these. So averaging over w in Q' gives a w0 with at most 64 mu (|C_k| + |C_k'|) / (eps |C_k'|) label-disjoint partners. - rho_k is not in L_w0, for two reasons: rho_k != rho_k', and the defining condition of Q'. So for every non-partner x, the shared value is one of |xa1|, |xa2|, and it lies in L_w0. -

Hence C_k is contained in S(a1, W) u S(a2, W) u H(a1, a2) u Ex. Here W = L_w0 u D_T u {rho_k}, so |W| <= 7, and the SAME W serves both centres. - The bound ex/|C_k| = (64 mu/eps)(1/|C_k| + 1/|C_k'|) <= 128 mu/(eps M) = delta1 is correct.

Step 1(ii). C_k' \ Q is contained in Lam_k' by (1) applied to k'. Q \ Q' is contained in S(a1, rho_k) u S(a2, rho_k). Correct. Good_k is existential in (W, Ex). That is exactly what Step 3 consumes.

Step 2, summing over k. - Radii are distinct for distinct indices. So each point of C_k' \ Lam_k' lies on at most one sphere S(a1, rho_k) with k in J, and on at most one S(a2, rho_k). - Hence sum_{k in J} (c - u_k) <= 2c, where u_k is the count controlled by (ii). - This gives c(|J| - 2) <= sum u_k < |J| eps |C_k'|, so c < 3 eps |C_k'| when |J| >= 3. Correct.

Step 3, the two-centre lemma (the crux). - Take x in R_lam = C_j n S(a2, lam) and a1 with lam notin W(a1). - x cannot lie on S(a2, W(a1)). - x lies on H(a1, a2) iff |xa1| = lam. - So x notin Ex(a1) forces |xa1| to lie in W(a1) u {lam}. Correct. - B on R_lam x Z_lam (disjoint, j != l): each factor |x|^2 + (|a1|^2 - v(a1)^2) - 2 has rank <= D+2; padding keeps the zero set; rank B <= (D+2)^{w+1}; nonzeros per column <= delta|C_j| <= |R_lam|/(16(D+2)^{w+1}). GL applies (real matrices, column-sparse form). - RL with k = w + 1: column functions v_1(a1), ..., v_w(a1), plus the constant lam. - |R'''| >= 2^{-L_{w+1}} (theta_abs - 1)/2 >= s' + 1, using w + 1 <= 8 and L nondecreasing in k. So OR forces |Z'''| <= s' and |Z_lam| <= 2^{L_{w+1}+1} s' + 1 <= e_{w+1}. Correct. - Pigeonhole: |W(a1)| <= w, so one of any w + 1 heavy radii avoids W(a1). Correct. - "Few" branch: HR with V = C_j (a2 in C_m, so a2 notin V; theta >= tau_min) leaves <= Gamma theta points of C_j off the <= w heavy a2-spheres, and Gamma theta <= |C_j|/(16(D+2)^w) since Gamma theta_abs <= M/(16(D+2)^8). Correct.

Exact form of HR used. HR is used with a = a2 in X, an arbitrary V subset of X \ {a} (namely V = C_j), and theta in [theta_abs, |C_j|]. Only "points of V in a-classes that meet V in < theta points number <= Gamma theta" is needed. See section 2 for why this holds.

Step 4. Rows C_j, columns F in C_m (j != m); rank <= (D+2)^w, column density <= 1/(16(D+2)^w); GL, RL (k = w), OR with |R'''| >= 2^{-L_w}(M-1)/2 > s'. |F| <= e_w. Correct (but see fix C on the definition of "w-few").

Corollary and its applications. - Centres a2 in F contribute <= |C_l| each. Any other a2 contributes <= (w+1) e_{w+1} by Step 3. - Arithmetic for (a): delta1 = 128 * 768 Gamma (D+2)^9 mu / M <= 1/(256 Gamma (D+2)^15) iff M >= 3 * 2^23 Gamma^2 (D+2)^24 mu. Correct. - Arithmetic for (b): 3 eps = 1/(256 Gamma (D+2)^9) = theta_rel(4) / (16 (D+2)^5), with equality. Correct. - Cov_j is a cover with W(a1) = {rho_l, rho_m, |a1a2|, rho_j}, so |W| = 4. It depends on a1, which is allowed. Radii in W(a1) for every a1 (rho_l, rho_m, rho_j) give Z_lam = {}, which the pigeonhole handles.

Step 5. - On V_lm, T is rainbow. At most one k *has rho_k* = |a1a2|. Every pair k != k' in I0 is valid. - If g <= |I0| - 4, then every k' in I0 has |J| >= |I0| - 1 - g >= 3. So Cov_k' holds for all k'. Hence g + #Cov >= |I0| - 3 >= K - 6. Correct. - Summing: (K - 6)|V_lm| <= (K - 2)[(e_7 + e_4)|C_l| + (8 e_8 + 5 e_5)|C_m|] <= 13 (K - 2) e_8 (|C_l| + |C_m|). With (K - 2)/(K - 6) <= 5/3 this gives |V_lm| <= 26 e_8 (|C_l| + |C_m|). - The file's display "(K-6)|V_lm| <= ... <= 2 * 13 e_8 (...)" drops the factor (K - 6). Typo (fix B). - Density <= 52 e_8 / M <= 1/(16 (D+2)^2) needs M >= 832 (D+2)^2 e_8. The proof has 960. Then GL, RL with 2 constant types, and OR with both sides > s'. Correct.

Quantifiers and constants. - Gamma, Q, tau_min, theta_abs, eps, delta1, M depend only on D and s' = s'_5(floor(D/2)); not on N, b or V. M = 2^{O(sqrt D log D)}(s'+1) (Gamma = D^{O(1)}, exponent ~ 340; mu carries 2^{L_9}, tau_min 2^{L_13} Q^{1/2}). - No circularity: HR uses only R+ levels r >= 1 = r0 (n = 5), which need no E1 (only G1, S2, GL, LR, OR, Lemma H, Caro-Wei). E1(5) uses HR, G1; Theorem R uses E1(5), G1. Induction is on D only. - Every use of |C_i| >= M is covered by the four terms

of M. - Optional remark: the count works already for K >= 7, where |I0| >= 4, if the constant 960 is replaced by about 2100. So "11" could be "6". This is not needed.

2. HR and R+ (the dependency E1(5) leans on)

HR is the statement inside the proof of Lemma HH (round12/03).

Construction. Put W_(a) := {x in V : |V n S(a, |ax|)| < theta}. Run levels 1-3 of R+ with tau_0 := theta and tau_r = Q^r theta.

Checks. - (i) The level-1 invariant ("classes of a meet W_(a) in < tau_0") holds by construction. {a, x} is trivially rainbow, and a is not in V. - (ii) Case B at level 1: x with F_b(x) != 0 is either in W_{(a,b)} or on S(a, |ab|), which is light by the invariant. The invariant is inherited by s_2 = b, since radii in R_b are excluded by rainbowness and radii in Big_b by definition. - (iii) Step A at levels 1 and 2, for n = 5, re-derived: - T = {b, a', s}, with s in S' a single entry of S. T is rainbow because Delta_b(a') != 0 and a' is not on S(s, |bs|). Also rho_i, rho_j are not in D_T. - Non-generic points of C_i lie on 4 spheres about a', on H(a', s), or in 4 light s-classes. - There are exactly 5 = n(n - 4) s-coincidence types in S2. With the F1 threshold theta_r = Q^{-1/2} tau_r: eta <= 10 Q^{-1/2} + 128 mu Q^{1/2} / tau_r <= 2^{-20} (D+2)^{-52}. - The four-case analysis gives at most 7 <= k' = 13 types: constants |bs|, rho_i, rho_j; column functions |a'b|, |a's|, t*(a'); row function |xs|. - GL, LR and OR then give |A_0| <= e_A. - (iv) Base level 3: the G1 degree form for Y = W_S, with degrees < 3 tau_2. - (v) No step uses points outside V except those of S and T; G1 is about all of X.

So m_1(theta) <= P_1 (2 max(m_2, e_A) + 22 theta + 1). For theta >= tau_min (which is >= e_A and >= mu) this gives m_1(theta) <= P_1 (202 P_2 Q^2 + 23) theta =: Gamma theta, and Gamma = D^{O(1)}.

Round12's "Gamma <= P_1 P_2 200 Q^2" is slightly off in the lower-order term. Defining Gamma as this explicit bound, or as sup_theta m_1(theta)/theta, fixes it (fix D). HR is therefore correct with uniform Gamma and tau_min, for every a in X, every V subset of X \ {a}, and every theta >= tau_min.

3. The chain to f_d(5)

Theorem R (round12/01), re-checked: - The claim (rainbow extension, or one of r^2 light classes). - The base G1 degree form: |Y| <= (4/3)(4^{n-2} mu + 1). - Caro-Wei plus the symmetric Lemma H.

With K = 11 and Typ = X (by E1(5)): - e_0 = 11, e_1 = 13, e_2 = 16, and m_3 = max(36 M, 128 mu + 4). - N <= m_0 <= (D+2)^{O(1)} (M + mu) = 2^{O(sqrt D log D)} (s'_5(floor(D/2)) + 1).

Recursion. We get s'_5(D) <= 2^{C sqrt D log(D+2)} (s'_5(floor(D/2)) + 1) for every D >= 1, with s'_5(0) = 1. Summing over the at most log2 D + 1 halvings gives log s'_5(D) = O(sqrt D log D). So f_d(5) = s'_5(d) + 1 = 2^{O(sqrt d log d)}.

Thresholds enter only through OR (smaller side has affine dimension <= floor(D/2)), and L_k with k <= 13 is O(sqrt D log D).

4. Adversarial examples

Families and tests. Rainbow-5-free families (at most 9 distances, or {b} u Y with Y at most 5-distance), exact integer arithmetic: J(9,4), J(12,6), J(7,2) x J(6,2) with b in X; {b} u J(10,3), {b} u J(13,2), {b} u J(9,5), {b} u (J(8,2) x simplex) with b a weight vector (17-35 classes at b); three greedy grid sets. Tests: T0 rainbow-5-freeness (sampling); T1 (1) both directions; T2 Step 1(i) non-partner cover for every w0 in

Q'; T3 (ii); T4 exact Step 2 inequality; T5 Step 3 slice implication; T6 G1 type list. About 150,000 checks, 0 violations; both Step 1 branches occur.

Structural consistency. - Relative 1-label hierarchies (rainbow-5-free, round9/02) have at most one big class per centre by Theorem D_q; E1(5) reaches the same end-state (V_lm is everything; GL, RL, OR contradict). Consistent. - {b} u Y with Y few-distance: b can have arbitrarily many classes, but all of size <= |Y|. This is polynomial in D, far below M >= 2^{L_9} s'_5(D/2) >= 2^{L_9} C(D/2 + 1, 9). This shows that E1 with a polynomial threshold would be FALSE (for example {b} u J(D+1, 5) gives about 5q classes of size about D^5/(600q)). So the super-polynomial factor in M is essential, and the proof does carry it. - Round12's (C2) configuration is excluded by E1(5) and is very probably vacuous (|Y| <= s'_4(D) likely < M).

Limitation (honest). Every known rainbow-5-free family has size polynomial in D, at most about 2^9 s'_5(D/2). Classes of size >= M can therefore never be realised, and the conclusions of Steps 3-5 cannot be tested numerically. Only the threshold-free inclusions were tested, and the threshold arithmetic was checked by hand.

5. Required fixes (all expository; none changes the mathematics)

- **A. State and prove HR as a lemma** in round13/01, or as an addendum to round12/03. It should include:
- the definition of W_(a) relative to V;
- the level-1 invariant;
- the remark that levels >= 1 = r0 need no E1;
- the explicit Gamma = P_1 (202 P_2 Q^2 + 23), or "Gamma := sup m_1(theta)/theta", with the bound m_1(theta) <= Gamma theta for theta >= tau_min. At present HR is assembled from "(*) with ground set V + dyadic regularity + proof of HH". It is correct, but it is cited rather than proved.
- **B. Step 5 display.** Write |V_lm| <= ((K-2)/(K-6)) * 13 e_8 (|C_l| + |C_m|) <= 26 e_8 (|C_l| + |C_m|).
- **C. Define "w-few" once:** "all but |C_j| / (16 (D+2)^w) points of C_j lie on at most w spheres about a2". Step 3 produces this property and Step 4 consumes it.
- **D. Gamma constant** in round12/03 (see section 2). Cosmetic.
- **E. Citation labels.** "referee12 item 4" (self-check) should point to round12/REFEREE_R.md.
- **F. One sentence:** (1) and (ii) need validity; Good_j, Cov_j are properties of arbitrary pairs, counted over all of C_l x C_m.
- **G. Status line.** "PROVED (candidate)" is appropriate. It should say: conditional on LR (new and unpublished; referee12 judged it correct) and on R+ Step A with fix F1 (REFEREE_R). If either fails, f_d(5) fails with it, but the deduction E1(5) => f_d(5) still stands. I recommend an external expert check of LR before any public claim.

6. Final verdict

- **Theorem E1(5): correct with fixable gaps.** The gaps are expository: A (state HR), B (display typo), C (one definition), and D-F (cosmetic). I found no mathematical error.
- The new ideas are sound: the w0-in-Q' covering with one shared radius set, the summing over k, and the two-centre slicing with HR. Every GL application really is in covering mode. Densities are <= 1/(16 (D+2)^{w+1}), 1/(16 (D+2)^w) and 1/(16 (D+2)^2), each with the matching rank bound.

- **n = 5 conclusion, f_d(5) = 2^{O(sqrt d log d)}: correct, conditional on Theorem LR (with G1) and R+ Step A (after F1).** Both were refereed earlier, and I re-derived the n = 5 instance of the latter in outline. Theorem R with K = 11 and the halving recursion check out.

Appendix M. Referee report on Theorem LR and n = 4 (referee 12)

Source file: e1088/referee12/REFEREE_REPORT.md

Referee report 12 (2026-10-19): Theorem LR, RQ-list, and f_d(4) = 2^{O(sqrt d log d)}

Referee brief: try to break (1) Theorem LR (round11/01) together with Lemma R and GL (round9/01); (2) RQ-list and its novelty relative to Singer-Sudan (SS, arXiv:2101.09592); (3) the assembled proof round11/n4_full_proof.md. Working notes: notes1_LR_RQlist.md, notes2_n4_lit_tests.md. Code: code/. Logs: *.log in this folder.

Summary of verdicts

claim	verdict
(1) Theorem LR (with Lemma R, GL, Lemmas A, B, NF)	**correct** (editorial remarks only)
(2) RQ-list, 2^{-O(k^3 sqrt d log d)} for k-listable rank-d matrices	**correct**; it does improve the bound SS call best known for parallel 3-partitions; I found no prior statement in print
(3) f_d(4) = 2^{O(sqrt d log d)} (n4_full_proof.md)	**correct** given (1) (editorial fixes only); still unrefereed by humans

I tried hard and could not find a mathematical error. Everything below was re-derived by hand. The numerical tests are sanity checks of the finite, deterministic parts. They cannot test the asymptotic claims, because every threshold is astronomically large.

(1) Theorem LR

Lemma R (margin rounding). Re-derived in full. - John position. Apply John to K0 = conv{±u_x} and normalise so that s^{-1/4}B ⊆ K0 ⊆ s^{1/4}B. Then |u_x| ≤ s^{1/4}. The support function of K0 in direction v_w is max_x|| ≤ K, which gives |v_w| ≤ K s^{1/4}. This is the standard γ2(N) ≤ sqrt(rank)·‖N‖∞ factorisation. - Margin. Since |N| ≥ 1/2, |cos(u_x, v_w)| ≥ b := 1/(2K sqrt s), with the sign of N. This answers point (a) of the brief. - Sheppard. The probability is 1/4 + arcsin(a)/(2π). It is at least 1/4 for a ≥ 0 and at most (1/4)(1 − 2b/π) for a ≤ −b, because arcsin b ≥ b. - Expectation. E|R∩Q−| ≤ mn·4^{-T}·exp(−2bT/π) ≤ (δ²/2)·mn·4^{-T} holds exactly when T ≥ πK sqrt(s)·ln(2/δ²). - Averaging. The averaging step is correct. So T = O(K sqrt(s) log(1/δ)) and the rectangle bound (δ/2)·4^{-T} per side are right (point (b)). - Numerics. MVEE-computed John positions on k-listable instances gave max|u|/s^{1/4} ≤ 1.0002 (the MVEE tolerance) and max|v|/(K s^{1/4}) ≤ 1. The margin ratio was always ≥ 1. At the stated T, E|R∩Q−|/E|R∩Q+| was ≤ 1e-19, far below δ/2. Sheppard's formula matched Monte Carlo to 1e-3.

GL (cleaning). Re-derived: 1. Delete rows and columns of degree > n/(4R') and > m/(4R'), respectively. 2. Take a maximal induced matching F; it has |F| ≤ rank, because it is a nonsingular diagonal minor. 3. Delete the neighbours of F. Maximality leaves no edge.

At least half the rows and half the columns survive. Integer rounding only helps: a count below 1 is 0, so the blocks stay nonempty. Tests on adversarial sparse low-rank matrices: a zero block was always found, retaining at least 0.777 of each side.

Lemmas A, B. All cases were checked. The costs (4,2) and (4,4) are correct. In case (c), D+E lies in (s/2, 8s), which gives the factor 16.

Lemma NF. - NF1. The pair rules give a constant strict sign for every surviving pair. A pair that ties on the kept part is deleted, and coverage is preserved. - NF2. Comparability holds with factor 16. - NF3. The winner argument is right, including case (c) with s < *s versus s* ≥ s and the degenerate cases. - Cost. The total cost is 2^{O(k²)}.

LR induction. - Rescaling (point (c)). N = (M − z)/σ with z = φ(x) + θ(w) and σ a basic quantity, so rank N ≤ r+2. The bound 1/2 ≤ |N| ≤ 256(k'−1) follows from g_{j*} ≥ *Q_{j*}* = σ and S ≤ 256(k'−1)Q_{j*}. *Under NF1, sign N records exactly which side of the j* gap an entry's unique type lies on. - Minority support (point (d)). Under NF1 the minority entries are exactly supp(∘_{τ∈S_maj}(M − τ)). Coincidences between types at isolated entries cannot survive NF, because NF either separates the two types strictly or deletes one of them. The rank is at most (r+1)^{|S_maj|} and δ = 1/(16(r+1)^{k−1}) matches GL. - Type removal (point (e)). After GL every entry of the half rectangle has a type in S_maj. S_min is nonempty because the split at j* is proper. So all minority types are removed everywhere on the rectangle, and the type count drops by at least 1. - Loss. Each round loses O(k²) + log(2/δ) + 2T = O(k²·sqrt(r)·log r) bits; with at most k−1 rounds the total is O(k³·sqrt(r)·log r).

Independent tests (code/test_nf.py). I wrote my own implementation of A, B, NF (with a faithful NF3) and one LR round. It ran on 1600 adversarial block-structured instances: accumulating labels κ ± 2^{−j} with j up to 40, ties a(x) = b(w) on blocks, magnitudes 2^{±40}, and k ≤ 5 types of mixed kinds. None of NF1–NF3, the |N| bounds, sign(N) = type split, supp(H) = minority, or rank H ≤ (r+1)^{|S_maj|} was ever violated.

Editorial remarks (optional). - (i) State explicitly that a pair (row function, constant) uses the row–row rule. "Constants count as both" implies this, but the pair rule only mentions row/row, column/column and row/column. - (ii) In Lemma R, "rank s" can be relaxed to "rank ≤ s" (T is monotone in s). - (iii) The sharper bound |N| ≤ k − 3/2 holds in RQ-list. This is harmless.

(2) RQ-list and the Singer–Sudan remark

Correctness. 1. Normalise each non-constant column at its own largest gap: N = (M − 1θ^{T})·diag(1/g). Then rank N ≤ r+1 and 1/2 ≤ |N| ≤ k − 3/2, and sign N_ij tells which side of column j's gap the entry is on. 2. Lemma R with δ = 1/(16 R_k) applies. 3. The wrong-side indicator is B = Σ_{t<k} M^{∘t} ·diag(c_t), using column-dependent Lagrange polynomials. Its rank is at most C(r+k−1, k−1), because rank M^{∘t} ≤ C(r+t−1, t). Tests confirmed rank(B) ≤ this bound in every case. 4. GL then leaves every column with values on one side of its gap, so at most k−1 values remain.

The induction is sound. There is no error. The only property of binarity that Lovett/Rothvoss use is the margin. Per-column rescaling at the largest gap supplies a margin with aspect ratio ≤ k, and the cost of the rank blow-up of the cleaning indicator is only logarithmic, because the rounding depends on log(1/δ).

Does it improve SS? Yes, as stated. - Exact translation. SS Prop. 2.3: a parallel k-partitioned configuration in R^d gives Mat(P,H) with entries , which has rank ≤ d and is k-listable. A column-constant S × T block gives a complete bipartite graph with |S|·|T| ≥ nm·2^{-2L}/k. - So for every fixed k, RQ-list gives rs(P,H) ≥ mn·2^{-O(k³ sqrt(d) log d)}. - SS (v5, p. 5–6) say the best known bound for

parallel 3-partitions is 2^{-Õ(d)}. They also say a 2^{-O(sqrt d)} bound is known "only" via binarity. RQ-list supersedes both remarks. - No contradiction. RQ-list does not touch SS's equivalence of Conjecture 2.6 with log-rank (that would need 2^{-polylog d}). It also does not prove SS Conjecture 2.7 (non-parallel partitions), and the authors do not claim it.

Is it already known? Searched: - HMST arXiv:2411.13510, "Disjoint pairs in set systems and combinatorics of low rank matrices". Thm 1.12: zero submatrix for nonnegative matrices with no entries in (0,1). Thm 1.13: nonnegative integer entries with average ≤ t give 2^{-O(t sqrt r)}. Appendix: t distinct entries with t ≥ r^{1+c} give 2^{-Θ(r log t)}, sharp (Construction 4). - Sudakov–Tomon arXiv:2311.18524 (binary only); Rothvoss arXiv:1409.6366 (±1 only); Lovett arXiv:1306.1877. - Every paper Semantic Scholar lists as citing SS, among them Hambardzumyan–Lovett–Shirley 2510.02583 and Song 2608.01812. Also web searches for "k-listable", "parallel partition" and "few distinct entries per column".

None of these states a 2^{-Õ(sqrt d)} bound for k-listable or q-ary real matrices. HMST remark that "the assumption that the matrix is binary is not crucial", but they prove only the integer and average-based version. HMST Construction 4 is consistent with RQ-list: there q ≈ r, where RQ's bound is vacuous.

Conclusion: RQ-list appears to be a correct, elementary, and (as far as I can find) unpublished observation. Since it follows from Rothvoss's proof in a few lines, experts may regard it as folklore. Before any public novelty claim, ask SS or Lovett. A full-text search of all post-2022 citing papers was not possible, because Semantic Scholar's citation list is incomplete (it omits 2411.13510).

(3) n4_full_proof.md

I re-derived every step (details in notes2). - **Step 1.** For x ∈ P and w ∈ Q, the five numbers c0, ρ(x), α(x), ρ(w), α(w) are pairwise distinct. Rainbow- 4-freeness therefore forces |xw| to equal one of them, so the k = 5 covering is valid. The rank is ≤ D+2. - A constant or column type makes every w ∈ C equidistant from R. - A ρ(x)² type gives R ⊆ H(o,w), and α(x)² gives R ⊆ H(a,w). Both lead to (OR) with p = o or p = a. - Then min(|P|,|Q|) ≤ 2^L·s' ≤ μ, and the degree form is right. - Exact test (code/test_geom.py): the covering was verified on all 1824 radius-disjoint pairs in greedy rainbow-4-free grid subsets of R² and R³. Every single-type 2×2 block found had orthogonal directions. - **Step 2.** - Lemma 4.4 is correct: m = min(|Z|, ⌊τ^{-1/2}⌋) > 1.2·C(D+2,2) and 5m/6 > C(D+2,2). - Blokhuis's C(d+2,2) bound for isosceles sets is cited correctly; I confirmed it via Ionin, EJC 16 (2009) R141. - The averaging gives τ0(|Lt|−2), and the degrees satisfy deg < 2βN ≤ |Y|/4. - **Step 3.** - Prop 4.6: K_i' ⊆ Y(o,a), and c0 ≠ ρ_i because a ∉ K_i. - Consequence (one sphere carries ≥ (1−ε) and ≥ (1−2ε)): the averaging is correct. - Prop 4.7: 16μ/ε ≤ εM. B is a degree-4 polynomial on the sphere with rank ≤ C(D+4,4), and ε matches GL. RL with k = 4 and |R| > s' gives |A| ≤ μ3. - Prop 4.8: the thresholds are right; for k = 1, 32(D+2)(ε1·N + 3) ≤ N holds iff N ≥ 192(D+2). - T: for o ∈ T there is a unique class of size ≥ M, and |Z_o| > ε1·N. - **Step 4.** - (a) Degrees in Y **are < 2M. This includes v = ρ1 occurring as an α-value, where the class of a has radius ≠ t1, hence is small. - (b) Crossing radius sets meet only if α(x) = ρ(w), and ρ(w) ≠ t1 because w ∈ Y. min(p,q) ≤ M + 16μ: I checked both p ≤ q and p > q. - (c) The count in m3 is right. -** Step 5. **rank F ≤ D+1, and the support is exactly the off-bisector set. A diagonal entry is −c0² ≠ 0. Charging gives ≤ (2s'+1)|T| pairs. -** Step 6. **The count m3' is right. rank G ≤ (D+1)(D+2) by (HP). The diagonal entry c0²λ² ≠ 0 makes the index sets disjoint. RL with rows 2-listable gives |B(o)| ≤ μ7. -** Step 7. **- The pair cases are exhaustive: (λ(o) ≠ λ(a), c0 not a label) goes to Steps 4/5; (λ(o) = λ(a) = λ, c0 ≠ λ) goes to Step 6; in every other case the pair is killed. - Edge count: (2s'+1)|T| + μ7|T|/2, because the B-relation is symmetric. So the average degree is ≤ 4s' + 2 + μ7, as written. - Caro–Wei is applied correctly. For Lemma H, F∘F^T is diagonal with a nonzero diagonal, so |I| ≤ (D+2)². -** Recursion. **- Every threshold in (T2)–(T6) is an N-free lower bound of the form poly(D)·2^{O(L5+L4)}·(s'+1). The only N-**

dependent quantity, e = ε1·N + 1 in Prop 4.8 with k = 1, is handled correctly. - So s'(D) ≤ A(D)·(s'(⌊D/2⌋) + 1) with log A(D) = O(sqrt(D)·log D). - Summing over the ≤ log2D + 1 levels (s'(0) = 1) gives O(sqrt(D)·log D), because Σ sqrt(D/2^i) = O(sqrt D). - There is no circularity: the thresholds use only s'(⌊D/2⌋). - (OR) is always applied to subsets of X, so both sides are rainbow-4-free. Orthogonal direction spaces force the smaller side to have affine dimension ≤ ⌊D/2⌋. - Consistency.** - Erdős #1088 is listed as open and gives no subexponential bound for any n ≥ 4, so the result would be new. - Known rainbow-4-free sets are polynomial (for example 5-distance sets), so there is no conflict. - Replacing LR by the trivial bound recovers 2^{O(d)}.

Editorial fixes for (3) (none mathematical). - (i) Step 1: "every Y in Y(o,a)" should read "every Y ⊆ Y(o,a)". Steps 4 and 6 use subsets. - (ii) Step 2: say that "isosceles" includes degenerate (collinear) triples. Blokhuis's bound is purely metric, so this is fine. - (iii) Step 7: say that the B-relation is symmetric; that is why μ7 enters dbar only once. - (iv) Prop 4.8: note that |H'| ≤ N/2 gives |X1| ≥ N/2. - (v) State once that every rectangle lemma is applied with integer floors, and that 2^{-L}|P| < 1 makes the size claim trivial.

Residual risks (honest caveats)

- No human expert has checked any of this. My check is a careful but single reading plus finite tests.
- LR is the single load-bearing new ingredient for (3). If it fell, (3) would revert to round 10's conditional theorem. I found no weakness in it.
- The constants are enormous: K = 256(k−1), and the thresholds are about D^{8}·2^{O(10^6 sqrt(D) log D)}. Only the shape 2^{O(sqrt d log d)} is claimed. That is correct but far from the polynomial lower bounds.
- Novelty of RQ-list and of the f_d(4) bound should be confirmed by domain experts before any public claim.